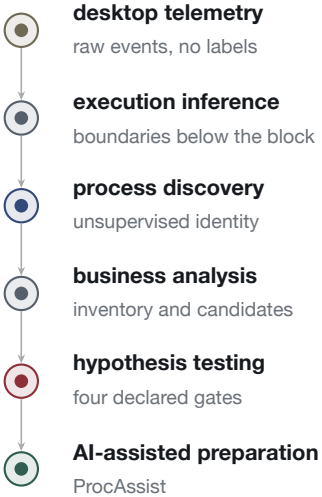
FORWARD-DEPLOYED ENGINEER / AI ENGINEER ASSIGNMENT • SEVEN-DAY ENGAGEMENT

Inferring Business-Process Executions from Desktop Telemetry

and Building a Document-Grounded AI Assistant for Back-Office Decisions

Final Technical Report



SCOPE

From raw keystroke-level desktop logs, with no process labels, to a running human-in-the-loop system that prepares back-office approvals against the regulations that govern them.

DELIVERABLES

1. executions recovered from telemetry, consistently labelled
2. a prioritised automation analysis of the production dataset
3. **ProcAssist** — a working automation tool
4. this report, the repository and its history

CLIENT CONTEXT

A Japanese company whose HR, Finance and Logistics back offices process routine paperwork across several internal systems.

EVIDENCE BASE

63 annotated sessions (162,768 events) • 15 production sessions (20,477 events), no ground truth.

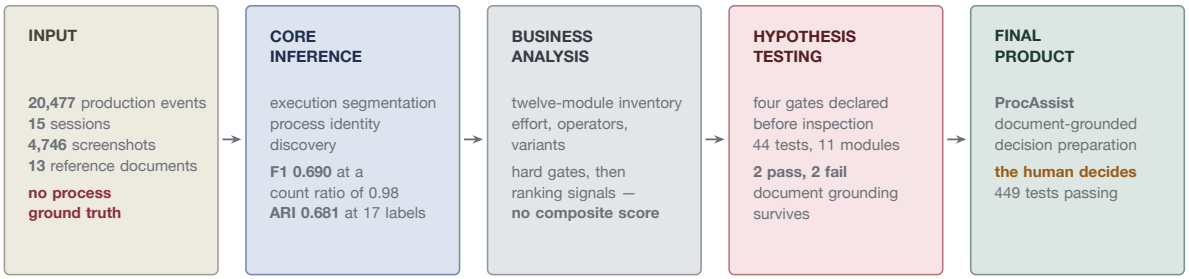
DOCUMENT

Typeset from FINAL_REPORT.md. Build: latexmk -xelatex main.tex. September 19, 2026.

Executive summary

The client's desktop agent records every keystroke, click and application switch from back-office staff. Nothing in that stream says *"this person is now processing an expense claim."* The logs had accumulated unread. Management wanted a prioritised automation proposal and a working demonstration.

PROJECT AT A GLANCE



Each stage exists because the stage after it cannot be trusted without it. Segmentation is instrumental: it exists so the analysis can be trusted, and the analysis exists so the automation target is the right one.

SIX FINDINGS THAT DECIDED THE ARCHITECTURE

1 | EXECUTION BOUNDARIES SIT *INSIDE* ACTIVITY BLOCKS

A segmenter restricted to activity-block edges can nominate only 59.9% of true boundaries. That was nearly read as a ceiling on the representation. It was not: adding an event-level candidate layer *inside* blocks took reachability to 88.0% against a measured ceiling of 90.0%. Boundary F1 **0.690** at a 2-second tolerance, 2.67× the strongest of six baselines, at a segment-count ratio of **0.98**.

2 | PROCESS IDENTITY WAS RECOVERED WITHOUT LABELS

Production has no labels and the annotated dataset's process codes cannot be transferred. Encoding each execution along three views — where, how, what — and partitioning a k-nearest-neighbour similarity graph by community detection reached **ARI 0.681 / NMI 0.738** at 17 discovered labels, against 0.431 at 51 labels for a structural labeller. Homogeneity and completeness rise at every step, which is the check that the gain is not over-merging.

3 | A TECHNICAL ROUTE IS NOT A BUSINESS PROCESS

The analysis aggregated executions by a display name derived from the URL route — and three different business systems serve the route `/payroll-items`. Invoice approval, expense settlement and inventory management had been merged into one "process family". The segmentation layer never lost the distinction; the analysis layer destroyed it in one function. The analysis was re-performed from scratch: **42 claims re-audited, 16 held, 11 contradicted**.

4 | THE ATTRACTIVE HYPOTHESIS WAS REFUTED, ON PURPOSE

Four gates were declared before the numbers were inspected. Two passed: the task, record and outcome are observable, and a governing rule parses from a captured document. Two failed: the deciding variable is present on **1.0–9.1%** of records, and **225 of 225** observed dispositions are positive across 25,495,968 characters of captured text. With no negative outcome there is no supervision signal.

5 | ONE SURVIVOR, AND IT IS STRONG

Across 11 modules, operators open a governing document before disposing at rates of **0.66–1.00** against a permutation null — **supported in 10 of 11**, most at $p < 0.001$. And the outcome signal is not absent, it is in the wrong field: of 481 distinct typed comments, **26** carry "discrepancy found / requires attention / requires confirmation". The learnable target moved from *predict the decision* to *predict the annotation*.

6 | THE PRODUCT AUTOMATES PREPARATION, NOT JUDGEMENT

ProcAssist resolves the governing regulation, quotes the governing clause, evaluates the record against it, reports what is missing and drafts the operator's comment — **with the judgement slot empty**. On the HR expense queue it resolves 23 documents and stages 120 drafts in about half a second, producing **11** OK **/ 12** FLAG **/ 97** MISS and **0 dispositions taken**. On the invoice queue it prepares all 108 records and **refuses all 108**. The refusal is the feature.

The summary in full

Why it is hard. Office work is not contiguous — an operator abandons a case mid-way and returns later. The same process recurs dozens of times a day with different data. The same process does not always follow the same steps. Unrelated activity is interleaved throughout. And the production dataset has **no ground truth at all**, so every claim about it is an inference that has to be justified by other means.

What we built, in one line. A four-layer chain: executions inferred from raw events → recurring processes discovered without labels → the production dataset analysed into a prioritised candidate inventory → the strongest candidate's automation hypothesis tested against evidence → and a product built around the part of the work the evidence actually supported.

Where a language model was sized, and declined. A language model was sized three times in the inference layer and declined three times, *each time against a measured ceiling* — not on principle. Label merging: a perfect-oracle merge tops out at ARI 0.478, *below* the 0.583 the clustering already achieved. Boundary adjudication: perfect adjudication of the entire ambiguity budget is worth $+0.026$ F1, and $+0.004$ at the budget actually used. Missing semantics: a logistic regression over exactly the features the clusterer sees recovers the reference process code at **0.891 accuracy**, so the residual is an unsupervised-objective gap, not a semantics gap a language model could fill. The scoped, deliberately unbuilt next step — metric learning from bounded model-generated pair labels — puts a model where the measurement says the gap is.

Where retrieval sits, and how it was decided. Governing-document retrieval was measured before it was built. On 148 labelled pairs over 17 document labels, a declared (module, record type) → regulation lookup reaches **64.2%**, 95% CI 56.2%–71.5%, against 5.4% random and 52.7% for per-module majority. *A learned retriever must exceed 71.5% before this evidence could distinguish it from the table at all* — so the lookup ships as the default, with the bar registered in advance, and a local lexical retriever runs beside it in the product where a reviewer can watch it disagree. On the flagship record it does disagree, matching the record's surface vocabulary to the noisiest part of the capture rather than to the governing clause. That disagreement is pinned by a test, and the retriever is labelled UNEVALUATED on screen because the 148 pairs are not recoverable from the committed artifacts.

Impact, stated in the only unit the data supports. The capture environment compressed waiting times, so no absolute time or monetary claim can be made from this corpus, and none is made anywhere in this report or in the product. The measurable unit is **operator actions**. Measured over 20,477 events with Wilson intervals: on the HR expense queue, 10 foreground switches into a word processor across 113 dispositions (0.0885 per disposition, CI 0.049–0.155); on the invoice queue, 45 across 76 (0.592, CI 0.480–0.696); on the managerial-approval queue, 27 across 32 (0.844, CI 0.683–0.931). Typed comments ran 35–44 characters where the staged draft is 31. The product ships a calculator whose three inputs — seconds per action, annual volume, loaded cost — are **empty**, because those are the client's numbers, not ours. § 20 derives the scenario arithmetic transparently and marks every assumption.

> **The engineering thesis**
>
> AI is placed where the evidence supports inference and withheld where it supports only a guess. **Unsupervised learning** does the work no rule could do — recovering process identity from unlabelled executions. **Retrieval and clause localisation** ground each decision in the document that governs it. **Deterministic validation** carries the invariants, because a wrong write to a statutory record must fail loudly rather than plausibly. And the **irreversible judgement stays with the human**, with the instrumentation in place to move that boundary later on evidence rather than on opinion.

The engagement

Engagement	Forward-deployed engineering assignment. Client: a Japanese company whose HR, Finance and Logistics back offices process routine paperwork across several internal systems.
Duration	7 days, ∼55 hours of recorded and reconstructed effort.
Brief, in the client's words	*"Use these logs to tell us where automation would have the greatest impact on our operations. And show us something that actually works."*
Deliverables	(1) executions recovered from telemetry and consistently labelled; (2) a prioritised automation analysis of the production dataset; (3) a working automation tool; (4) this report, plus the full repository and its history.
Language	All screen text, process names, documents and operator comments are Japanese. Japanese source text is preserved verbatim throughout and glossed, never replaced.
Stack	Python 3.12 • NumPy / SciPy / scikit-learn / NetworkX / RapidFuzz / segeval for the inference layer • FastAPI + Pydantic v2 + Jinja2 + SQLite + vanilla JS for the product • **no vector database, no model API, no agent framework**.
Test suite	449 passing, 2 skipped (366 test functions, 79 of them for the delivered product).

How to read this report

In two minutes	the executive summary and Figure 1.
In fifteen	add §5 (what was measured), §10.5 (what was refuted) and §12 (what was built).
As an evaluator	§15 states where AI is used and why, §19 is every number with its population, and §21 is what could still be wrong.
To run it	§18: two commands to a running system and a ten-step verified walkthrough.

Reading the evidence labels. Every claim in this report carries the status of its evidence, and the labels are used consistently: MEASURED a number produced by a recorded experiment on a stated population; TESTED a property enforced by an automated test rather than scored; OBSERVED read directly from the captured data; INFERRED derived rather than observed; DECLARED a policy parameter chosen by us, not measured; NOT MEASURED named rather than estimated. Nothing in this report is estimated where it could have been measured, and nothing is presented as measured where it was not.

Colour is meaning, not decoration. The same encoding is used in every figure:
deterministic / classical · learned or inferential (AI) · data and evidence
human judgement · failure, refutation, uncertainty · the delivered path
Every hue also differs in lightness, so the report reads correctly in grayscale printing.

Contents

Executive summary i

The engagement, and how to read this report iv

1 Problem statement 1
- 1.1 What the raw data is 1
- 1.2 What a “business-process execution” is 1
- 1.3 Why low-level events are not enough on their own 1
- 1.4 Why repeated activity does not imply an automation opportunity 1
- 1.5 The end goal 2

2 Data and experimental setup 2
- 2.1 Dataset A — the measurement instrument 2
- 2.2 Dataset B — the production target 2
- 2.3 Telemetry limitations, stated up front 3

3 Overall methodology 4
- 3.1 The rationale, layer by layer 4

4 Layer 1 — inferring process executions from raw events 5
- 4.1 Telemetry normalisation 5
- 4.2 Activity context — a container, not a boundary unit 6
- 4.3 Baselines 6
- 4.4 Boundary analysis 7
- 4.5 Error taxonomy 8
- 4.6 Candidate generation 9
- 4.7 Segmentation 9
- 4.8 Process identity 10
- 4.9 Alternatives considered 11

5 Layer 1 — results, and what they cost to believe 11
- 5.1 Boundary placement 11
- 5.2 Process identity 12
- 5.3 Where a language model was sized, and why it was declined 13
- 5.4 The residual, named as a finding 13
- 5.5 Additional limitations of this layer 14
- 5.6 An operational failure worth recording 14
- 5.7 Key findings, and why they mattered downstream 15

6 Layer 2 — business-process analysis of the production dataset 15
- 6.1 The module inventory 15
- 6.2 The tension the analysis surfaced 16
- 6.3 What the analysis could not establish 17

7 Literature and industry review 17
- 7.1 Process and task mining from interaction logs 17
- 7.2 What makes automation candidates succeed 18
- 7.3 Prioritisation methodology 18
- 7.4 Autonomous agents on enterprise interfaces 19
- 7.5 The Japanese enterprise and public-sector context 19
- 7.6 Synthesis 20

8 The automation opportunity framework 20
- 8.1 Why not a single score 20
- 8.2 Hard gates — evidence sufficiency, applied before ranking 21
- 8.3 Ranking signals, kept separate 21
- 8.4 The candidate matrix 23

9 The correction that mattered 23
9.1 The candidates as first analysed 23
9.2 The uncomfortable observation 23
9.3 A technical route is not a business process 23
9.4 The correction was itself corrected 25
9.5 The hypothesis this produced 26
10 Layer 3 — testing the hypothesis before building on it 26
10.1 Why this layer exists 26
10.2 The evidence layer built to answer it 26
10.3 The four hypotheses, 44 tests 26
10.4 The outcome-variance sweep, and the error I made 28
10.5 The four-gate verdict 29
11 The pivot: from autonomous decision to AI-assisted preparation 29
11.1 Why this is a narrowing, not a failure 29
11.2 The safety argument for the boundary 30
12 ProcAssist — the delivered system 31
12.1 The end-to-end journey 31
12.2 Scope: a shared foundation, one module deep 32
12.3 What it deliberately does not do 34
13 High-level design 34
13.1 Component responsibilities 36
13.2 The definition file is the artifact that executes 36
13.3 Status vocabulary, and the absence of a score 37
13.4 Integration surfaces 37
14 Low-level design 37
14.1 Request lifecycle 37
14.2 Request and result schema 38
14.3 Endpoints 39
14.4 Definition format 39
14.5 Document representation and clause localisation 40
14.6 Validation logic, and one correction 40
14.7 Draft generation 42
14.8 Trace, persistence and configuration 42
14.9 Panel interaction and its invariants 43
14.10 Error handling 44
15 Where the intelligence is 44
15.1 Alternatives considered across the whole system 47
16 Retrieval and document grounding 48
16.1 The corpus and its identity problem 48
16.2 How much supervision exists 48
16.3 Baselines, and the registered bar 49
16.4 Why no vector database 49
16.5 The challenger, and the failure mode it exposed 49
16.6 Provenance 49
17 Human-in-the-loop design and safety 50
17.1 Where the boundary is 50
17.2 Risks, and what the architecture does about each 50
17.3 The honesty labels 51
18 Running the system 51
18.1 Prerequisites 51
18.2 Quick start 51
18.3 The full end-to-end run 51
18.4 The other three cases 52
18.5 Optional: watch the retriever lose 52
18.6 Read the audit trail 52
18.7 Tests 52
18.8 Reproducing the earlier layers 52
18.9 Troubleshooting 53

19 Evaluation 53
19.1 Layer 1 — execution inference (annotated dataset) 53
19.2 Layer 2 — analysis validation (production dataset) 54
19.3 Layer 3 — hypothesis testing (production dataset) 54
19.4 ProcAssist evaluation 54
20 Expected business impact 55
20.1 What can and cannot be claimed 55
20.2 Observed — the action budget 55
20.3 Counterfactual — what the prototype removes 56
20.4 Scenario arithmetic — the model, not a number 56
20.5 The impact that is not in the arithmetic 57
20.6 What would make impact measurable 57
21 Failure modes and limitations 57
22 Security, privacy and governance 58
23 What we would do next 59
23.1 Immediate production hardening 59
23.2 Additional evaluation 60
23.3 Broader coverage 60
23.4 Model and retrieval improvements, each with its trigger 60
23.5 Integration 60
24 Conclusion 60
25 References 61
A Dataset statistics 62
B Metric definitions 63
C Baseline comparison 64
D Process-discovery experiments 64
E Step 2 candidate analysis 64
F Hypothesis matrix 64
G API schema 65
H Example request and response 65
I Configuration 65
J End-to-end run 65
K Impact calculation methodology 66
L Supporting material in the repository 66

List of figures

1 From desktop telemetry to an AI-assisted business workflow . . . 4
2 Raw telemetry to executions . . . 5
3 An activity block is not a business-process execution . . . 6
4 Boundary F1 is not interpretable on its own . . . 7
5 The candidate space, not the representation, was the ceiling . . . 8
6 Where the missed boundaries actually went . . . 8
7 Candidate generation, and why it is a separate stage . . . 9
8 The candidate-space trade-off, as it was actually navigated . . . 10
9 Process identity is the unsupervised core of Layer 1 . . . 10
10 Four metrics, not one . . . 12
11 What perfect adjudication would be worth . . . 14
12 The production process inventory, and the tension in it . . . 16
13 How the literature informed the engineering . . . 18
14 Hard gates, ranking signals and open business questions — in that order, and never averaged . . . 20
15 Candidates placed by assessment, not by score . . . 22
16 The methodological lesson of the engagement . . . 24
17 One wrong key, not three partial views . . . 25
18 The four-gate test, and what survived it . . . 27
19 One hypothesis survives, and it is not the one that would have been convenient . . . 28
20 What was abandoned, and what replaced it . . . 30
21 ProcAssist in place . . . 31
22 The operator journey, with the two human steps drawn in the autonomy convention . . . 32
23 The starting state: the operator's own queue, unassisted . . . 33
24 The queue after preparation, with the decision boundary intact . . . 33
25 ProcAssist, as implemented . . . 35
26 One request, eight steps, two ways to refuse . . . 38
27 The result contract as a data model . . . 38
28 The same result object the panel renders, available over HTTP . . . 40
29 Every statement resolves to a source . . . 41
30 Deterministic validation, shown in three separable parts . . . 42
31 The draft, with the judgement slot empty . . . 43
32 Where the system stops . . . 43
33 When each status is produced . . . 44
34 Missing information is reported, never imputed . . . 45
35 The refusal is the feature . . . 45
36 The product boundary . . . 50

List of tables

1 The four properties of office work that rule out contiguity-based, content-keyed and fixed-path segmentation. . . . 1

2 The two datasets and their roles. Dataset A is the measurement instrument; Dataset B is the thing the client actually wants understood, and it carries no annotation of any kind. 2
3 Telemetry limitations and the design decision each one forced. The middle column is the measured extent, not an impression. 3
4 Six baselines and the delivered pipeline, evaluated identically. **The count-ratio column is the one that matters.** The strongest baseline achieves its F1 while emitting 2.83× as many segments as there are executions; the surface-change baseline emits 5.03×. Boundary F1 improves mechanically when a method cuts more often, so a headline F1 without a count ratio beside it is not interpretable. 7
5 Boundary-miss attribution and what each category caused. Categories are assigned per boundary and proven rather than assumed. 9
6 The three-view execution representation. Similarity combines a sequence channel over surface shapes with set-overlap channels and a profile distance; the weights are reported and swept ±20% rather than presented as discovered constants. 11
7 Alternatives for segmentation and identity, with the measurement that decided each. Oracle-k results on the annotated data are reported separately from unknown-k results on production, because conflating them inflates the apparent result. 11
8 Boundary placement across the full tolerance curve, each row paired with the reachability ceiling at the same tolerance. Results are reported at 2 s because the annotation leads the nearest retained event by ≈176 ms at the median — below 1 s the metric measures the annotation offset rather than the method. 12
9 Process identity, three arms on an identical segmentation. **Homogeneity and completeness both rise at every step**, which is the check that the improvement is not over-merging — ARI alone cannot distinguish "better clusters" from "fewer, larger clusters", and reporting it alone is how clustering results get oversold. 12
10 Why matched-subset identity metrics are not comparable across arms with different match rates. **Event ARI** — computed over all events rather than a matched subset — is the comparable metric, and it puts the pipeline ahead at 0.297 against 0.250. Two honest qualifications: the identity margin (1.19×) is far narrower than the 2.67× on boundary placement, and the dwell baseline reaches its 0.250 with 130 labels at a 2.83 count ratio, which is not a process inventory anyone can act on. 13
11 Three proposed roles for a language model in the inference layer, and the measurement that closed each. The third is decisive: if a linear model over the same features recovers the process code at 0.891, the residual is an *unsupervised-objective* gap, not a semantics gap a language model could fill. 13
12 Layer-1 findings and their downstream consequences. The sixth row is the bridge: Layer 1 had already established, on the annotated data, that UI-level evidence can be identical for different business purposes. Layer 2 then walked into exactly that trap on the production data — and recognised it because the limit had already been named. . . 15
13 The six principal modules of the twelve-module inventory. *Module purity* is the share of an execution's business activity belonging to its dominant module — the measure of how cleanly the segmentation isolated it. Time shares are **comparative only**: waiting times were compressed in the capture environment, so no absolute duration claim is possible. 16
14 The evidence matrix, abridged. Each row records what the signal is, whether it is directly observable, and the failure mode that stops it being trusted as business semantics. The route row is the one that later cost a full re-analysis. 17
15 Three prioritisation families examined. The method adopted takes the criteria from all three and **refuses the aggregation step**; § 8 explains why. 18
16 Every row is a research finding paired with the specific engineering decision it produced. This table, not the citation list, is what the literature review was for. 20

17 The five hard gates. Two modules were excluded on the first gate; one — purchase-order management — failed identity confidence at purity 0.69, and its exclusion is itself informative, because the contamination came from the five-way over-merge described in § 9. . . . 21
18 Eleven ranking signals and three that do not exist in this data. The observability column is why the framework refuses to aggregate: two of the signals are only partly observable and three are absent entirely, so a composite would silently weight a guess against a measurement. . . . 21
19 The candidate matrix. Two entries are worth reading as judgement rather than output: onboarding verification has the *best* identity purity (0.90), the lowest variance and the cleanest two-application footprint — and is marked *AI would be decoration*, because a deterministic rules engine covers it. . . . 23
20 Three-key triangulation over 3,011 extracted records. Only two patterns occur: all three agree (2,064 records), or **the URL dissents alone** (552 records). The screen header and the prefix never disagree with each other — and that asymmetry is the signature of one wrong key, not three partial views. . . . 25
21 44 tests over 11 modules. The evidence floor, the permutation count and the seed were all fixed before inspection. . . . 28
22 The four-gate verdict. G3 and G4 fail *compatibly*, which is what makes the refutation structural: if the deciding variable is absent from 91–99% of records, the operator is not deciding from the record. . . . 29
23 The five arguments for placing the boundary at reversibility. Where the evidence *does* license autonomy, the system takes it: it prepares 120 records unattended, without asking. The boundary is drawn from measurement, not from caution as a style. . . . 30
24 Three declared queues, each demonstrating a different status. Three rather than two is an evidence-forced decision, not padding: the corpus contains **exactly one** parsed threshold rule, and no record in the first two queues *lacks* an amount, so the missing-input status was unreachable without a third screen — and fabricating a gap to demonstrate a status was not an option. . . . 32
25 Seven capabilities deliberately withheld, each with the evidence that withheld it. This table is the product specification as much as the feature list is. . . . 34
26 Component responsibilities and, crucially, failure behaviour. The right-hand column is the design: every component has a defined way of refusing, and none of them degrades quietly. . . . 36
27 The four statuses. A single number would invite trading these apart — "0.7 ok" — when three of the four mean *stop and look* and only one means *the rule held*. . . . 37
28 Integration surfaces delivered and deliberately not delivered. The last row is the highest-value open question in the engagement, and the architecture is shaped to keep its answer available. . . . 37
29 Eight service endpoints and four server-rendered pages. Nothing in the request path reaches a model, and there is no endpoint that performs a disposition. . . . 39
30 Error handling. Every row is a way of refusing rather than a way of degrading — which is the same principle the inference layer learned the hard way in § 5. . . . 44
31 Mechanisms by layer. Learned components are used only where the available supervision and failure cost support them. . . . 46
32 Alternatives rejected, retained or deliberately deferred. . . . 47
33 The supervision available for governing-document retrieval. . . . 48
34 Governing-document retrieval baselines. The label-space choice is shown rather than hidden because it moves the result materially. . . . 49
35 Safety risks, evidence, mitigations and residual gaps. . . . 50

36 Ten-step verified walkthrough. 51
37 Cases covering the three non-FLAG statuses. 52
39 Layer 1 evaluation. 53
41 Layer 3 verdicts. The outcome sweep covered 25,495,968 characters, 20,477 events, 4,746 screenshot filenames and every status transition. 54
42 Product evaluation. Dashes are retained where evidence does not exist. 54
43 Observed document-switching and comment-length budget. Variation is itself a prioritisation signal. 56
44 Measured HR-expense counterfactual over 120 records. Rates remain ranges and are never converted to seconds without client inputs. 56
45 Inputs to the scenario calculator. Missing quantities are left visibly empty rather than estimated. 56
46 Limitations are reported with their mitigations and remaining gaps, not converted into a confidence score. 57
47 Security, privacy and governance position. 58
48 Dataset populations. 62
50 Process-discovery ablation. 64
51 Layer 3 hypothesis matrix. 64

1 Problem statement

1.1 What the raw data is

A desktop agent installed on staff machines records a chronological stream of low-level interactions: key presses, mouse clicks, scrolls, application and window switches, browser navigations, clipboard operations, form inputs, and periodic screenshots. Each event carries a timestamp, the foreground application, window title, and — for browser events — the page origin and route. Some events carry text extracted from the screen.

> **The whole of the input**
>
> There is **no case identifier, no transaction id, no process name**, and no marker of any kind saying where one piece of work ends and the next begins.

1.2 What a "business-process execution" is

One coherent pass through a business procedure for one case: opening a queue of pending items, selecting a record, checking it against whatever governs it, entering a comment, and dispositioning it. An execution is the unit a business person would call "one of those" — one expense claim processed, one invoice approved, one new hire verified.

The gap between the two definitions above is the whole of Layer 1. Recovering executions from events is the enabling capability; without it there is nothing to count, compare or automate.

1.3 Why low-level events are not enough on their own

Four properties of real office work defeat the obvious approaches.

Property	Why it defeats the obvious approach
Work is not contiguous	An operator begins a case, is interrupted, handles something else, and returns. A segmenter that assumes one execution occupies one unbroken span of time will fragment every interrupted case and merge every quick one.
The same process recurs, with different data	Dozens of cases pass through the same procedure in a day. The procedure is the same; the values, the vendor, the amount and the outcome differ. Any method keyed on the specific content of a case cannot recognise the next one.
The same process does not always follow the same steps	Which systems are visited and which items are checked depend on the case. An expense at ¥40,000 and one at ¥140,000 are the same process with different paths through it.
Unrelated activity is interleaved	Email, browsing, file management and machine setup are mixed into the stream and belong to no business process.

Table 1. The four properties of office work that rule out contiguity-based, content-keyed and fixed-path segmentation.

Additionally, the two signals that look most useful are both traps. **Recording chunks** — the agent's own buffering boundaries — are an artifact of how data was written to disk; treating them as process boundaries measures the recorder, not the work. **Temporal gaps** are weak: batching means an operator often moves from one case to the next with no perceptible pause, while a genuine mid-case pause can be longer than a between-case one.

1.4 Why repeated activity does not imply an automation opportunity

This is the distinction the engagement turns on, and it is three questions, not one.

1 | Is it automatable? Is the action sequence deterministic, the input structured, the interface stable, the variance across repetitions low?

2 | Is it worth automating? How often does it happen, how much aggregate effort does it consume, how many people does it affect, what will it cost to maintain?

3 | Is autonomous action appropriate? Can the decision be recovered from evidence? Is there an outcome signal to learn from and validate against? What is the cost of a wrong action, and is it reversible?

A task can be highly repetitive, cheap to automate, and still a bad target — because the repetition is the *easy* part and the judgement wrapped around it is the reason a human is there. § 8 develops this into the framework actually used, and § 10 is what happens when the third question is asked seriously.

1.5 The end goal

Business value, not segmentation accuracy. Segmentation is instrumental: it exists so that the analysis can be trusted, and the analysis exists so that the automation target is the right one. A perfect segmenter attached to the wrong automation proposal is worth nothing to the client.

2 Data and experimental setup

Two datasets with deliberately different roles.

Property	Dataset A	Dataset B
Role	engineering and validation	**production analysis target**
Sessions	63	15
Recording chunks	133	20
Events	162,768	20,477
Screenshots	34,563	4,746
Distinct machines	8	4
Process ground truth	**yes** — a per-session annotation file recording when each business process started and ended	**none**
Departments / applications	one set	**different departments, different processes, different applications**
Reference documents	—	13 (Word regulations, checklists and reference tables)

Table 2. The two datasets and their roles. Dataset A is the measurement instrument; Dataset B is the thing the client actually wants understood, and it carries no annotation of any kind.

2.1 Dataset A — the measurement instrument

Dataset A carries annotations, so every design choice in Layer 1 could be *measured* rather than argued: boundary placement against annotated start and end times, and process identity against annotated process codes. It is the only place in the engagement where a supervised number exists.

2.2 Dataset B — the production target

Dataset B is the data the client actually wants understood, and it has no annotations. It also comes from different departments performing different work with different applications, which forbids the most tempting shortcut: carrying anything dataset-specific across.

That prohibition was made structural rather than aspirational. **Transferable:** code, featurisation, the *rule* for estimating a parameter, the evaluation harness, domain-neutral prompts. **Not transferable:** Dataset A's process names and codes, its vocabulary, keyword lists, its cluster count, its memory records, its few-shot examples. The production pipeline is tested with the annotated dataset *absent from disk*, so a dependency on it fails the build rather than silently improving a result.

2.3 Telemetry limitations, stated up front

Every one of these shaped a design decision. None is a caveat added after the fact.

Limitation	Measured extent	Consequence
Values are redacted	field values are hashed, not persisted	any method needing the actual data of a case is impossible; the literature's standard automatability index, which requires values, had to be replaced by a redaction-safe structural proxy
Extracted screen text is sparse	present on 939 of 20,477 events (**4.59%**)	it is rich where it exists — full queue tables, detail panes, document bodies, outcome messages — but it bounds every measurement derived from it. This is an *evaluation* limit; the delivered product reads a live interface, not telemetry
The recorder's own OCR is unusable	renders 第1条(目的) as 第1条(白的), interleaved with word-processor ribbon text	optical character recognition was eliminated as an input on evidence, not on preference
Recording chunks split sessions	133 chunks over 63 sessions; 20 over 15	chunk edges are dissolved during normalisation and never used as process boundaries
Annotation lead	ground truth precedes the nearest retained event by ≈176 ms at the median	any tolerance below 1 second measures the annotation offset, not the method. This is why results are reported at a 2-second tolerance with the full tolerance curve alongside
One truncated session	its annotation extends far beyond its telemetry	scores F1 0.022 and remains in the reported mean, named as that session and never as a dataset-wide failure
Waiting times are compressed	stated by the data provider	**no absolute time claim is possible from this corpus.** Every duration is comparative only; the impact model counts actions, not seconds
Records recur across captures	324 of 898 records seen in more than one status	the product shows the first observation and discloses the rest rather than silently choosing
Consecutive same-process executions are untestable on A	0 of 1,689 consecutive annotated pairs share a process code	the behaviour could only be checked on the production data, where the frozen system emits 67 such pairs out of 168 (39.9%) — so it does split adjacent same-process work rather than merging it
Rare processes are untestable on A	all 15 reference codes appear $\geq$ 14 times among 1,332 matched executions	nothing can be said about long-tail process behaviour

Table 3. Telemetry limitations and the design decision each one forced. The middle column is the measured extent, not an impression.

Always state the ground-truth population

Dataset A's annotation is evidence, not truth, and it is imperfect. The populations are: raw 2,009 • canonical 1,819 • interrupted 98 • unscoreable 230 • **scoreable canonical 1,589**. Every accuracy figure in this report uses the scoreable canonical population. Candidate *reachability* is reported separately and is never presented as an accuracy number.

3 Overall methodology

The engagement is one dependency chain. Each layer exists because the one after it cannot be trusted without it.

STAGE	INPUT → OUTPUT	MAJOR METHOD	KEY EVIDENCE
RAW TELEMETRY	keystrokes, clicks, app switches, screenshots → one chronological stream	the desktop agent; nothing is added or discarded	**162,768** events on A **20,477** on B no case id anywhere
NORMALISATION	raw chunked stream → canonical surfaces	deterministic; chunk membership dissolved; *(origin, route)* or app + document identity	"destroyed by preprocessing" measured at **zero** 176 self-switch signals kept
EXECUTION INFERENCE	events + activity blocks → execution boundaries	candidate generation, fitted scorer, ambiguity gate, constrained global decoder	**F1 0.690** @ 2 s, count ratio **0.98**; reachability 59.9% → **88.0%** of a 90% ceiling
PROCESS IDENTITY	unlabelled executions → 17 recurring processes	where/how/what views, kNN similarity graph, community detection	**ARI 0.681 / NMI 0.738** vs 0.431 at 51 labels; homogeneity *and* completeness rise
DATASET B ANALYSIS	frozen executions → a twelve-module inventory	37 measured columns per module; three keys triangulated; screen-text validation	header ↔ prefix agree **2,904 / 2,904**; the URL dissents alone in all 552
AUTOMATION OPPORTUNITY	inventory → a ranked candidate set	hard gates, then separate ranking signals, then open business questions — **no score**	largest process by time is the *least* repeatable; no candidate scores well on both axes
HYPOTHESIS TESTING	the strongest candidate → a verdict on automating it	four gates declared first; permutation nulls, 10,000 draws, evidence floor $n \geq 8$	deciding variable on **1.0–9.1%**; **225/225** dispositions positive; documents opened first in **10 of 11**
PROCASSIST	a record on screen → prepared evidence + a draft	declared lookup, clause localisation, deterministic validation, template draft	**0 dispositions taken**; 23 documents + 120 drafts in ~ 0.5 s; **449 tests**

Figure 1. From desktop telemetry to an AI-assisted business workflow. The visual anchor of this report. Each stage names what it consumes, what it produces, the method it uses, and the measurement that justifies it. Reading down the right-hand column gives the evidence base of the whole engagement: nothing in the chain is asserted without a number behind it, and two of the eight stages exist purely to establish that the stage before them can be trusted.

3.1 The rationale, layer by layer

Normalisation removes artifacts of the recorder so that nothing downstream can accidentally learn them. **Activity context** compresses ∼20k events into a manageable number of interaction runs and gives each event a stable notion of "where am I" — but it is explicitly not assumed to delimit business work, and § 4.2 explains why that distinction turned out to be the single most consequential decision in Layer 1. **Candidate generation** separates *what positions can even be proposed* from *which are chosen*, so that recall failures can be attributed to the right stage. **Constrained segmentation** makes the output satisfy the structural invariants by construction rather than by hoping. **Representation and identity** are where the problem becomes a machine learning problem, because the production data has no labels. **Analysis** turns executions into a business inventory. **The candidate framework** separates automatability from value from safety. **Investigation and hypothesis testing** exist because the analysis produced a hypothesis attractive enough to be worth trying to falsify before spending the build budget on it. **The product** is what survived.

The reporting discipline that makes the chain auditable

Three rules are applied without exception, and they are what makes the numbers in this report comparable to each other. **(1) Every accuracy figure states its population. (2) Candidate reachability is never presented as an accuracy number** — it answers "can this position be proposed at all", which is a different question from "was it chosen". **(3) No boundary F1 appears without its segment-count ratio**, because boundary F1 improves mechanically when a method cuts more often.

4 Layer 1 — inferring process executions from raw events

The task

Given a stream of low-level desktop events, recover coherent business-process executions and label recurring ones consistently.

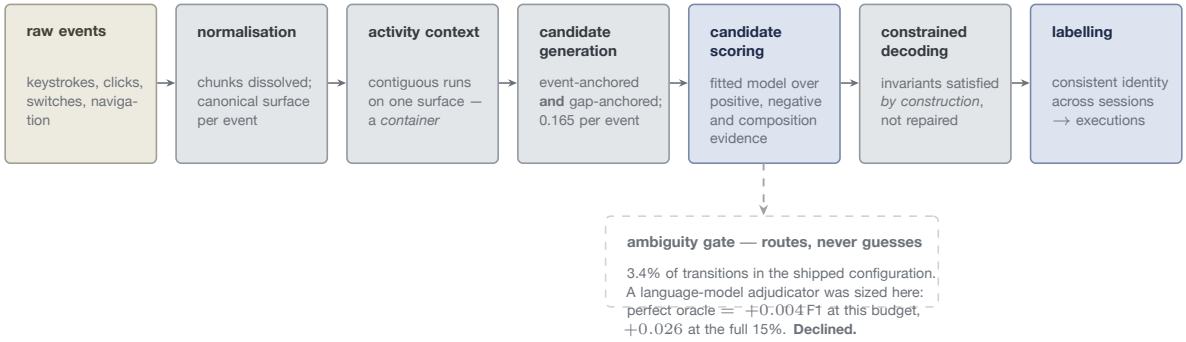


Figure 2. Raw telemetry to executions. Every stage on the critical path is deterministic except the fitted candidate scorer and the labeller. The ambiguity gate *routes* the residual rather than guessing at it; a language-model adjudicator was sized for that slot against the measured value of perfect oracle performance, and declined.

4.1 Telemetry normalisation

Events arrive per recording chunk. Normalisation dissolves chunk membership entirely — a session becomes one continuous stream — because chunk edges are a property of the agent's buffering and using them would measure the recorder. Provenance and correlation fields present in the raw events were examined and rejected as an identity mechanism: they look like a shortcut and are not one.

Each event is then assigned a **surface**: a canonical identifier for the thing the user was operating, defined only by user-operation layers. For a browser this is (origin, route); for a desktop application it is the application plus a normalised document identity. Two early defects were fixed here rather than compensated for downstream: browser window titles carried a suffix that fragmented otherwise-identical surfaces, and untitled windows produced a noise signature that polluted comparisons.

One decision in this layer proved unexpectedly important. The recorder emits app_switch events whose source and target are the *same* application. These look like artifacts and were nearly suppressed. Measurement showed that suppressing them would have destroyed **176 genuine document boundaries**, and they later turned out to be the highest-mass signal for boundaries occurring inside a single application context.

4.2 Activity context — a container, not a boundary unit

Contiguous runs of interaction within one surface are grouped into **activity blocks**. A block is a compression and context device: it gives every event a stable local identity and reduces ∼20k events to a tractable number of units.

The initial hypothesis was stronger than that — that blocks were natural execution units. A person works in a form, switches away, comes back; segment at block edges. It is an attractive assumption and it is wrong.

The measurement that broke it. The share of annotated boundaries that a block-edge-restricted segmenter could even *nominate* — its reachability — was 59.9% at a 2-second tolerance. That number was nearly read as a ceiling on the representation: evidence that normalisation had destroyed the information. A dedicated experiment then asked whether the information had survived, at event level and at field level, and found the "destroyed by preprocessing" category to be **zero on both**.

The information was present and unsurfaced. Real operators finish one case and begin the next without changing application, window or route — the only evidence is a submit, a field reset, a new record identifier. A block is therefore a **container**, and boundary inference has to operate below it.

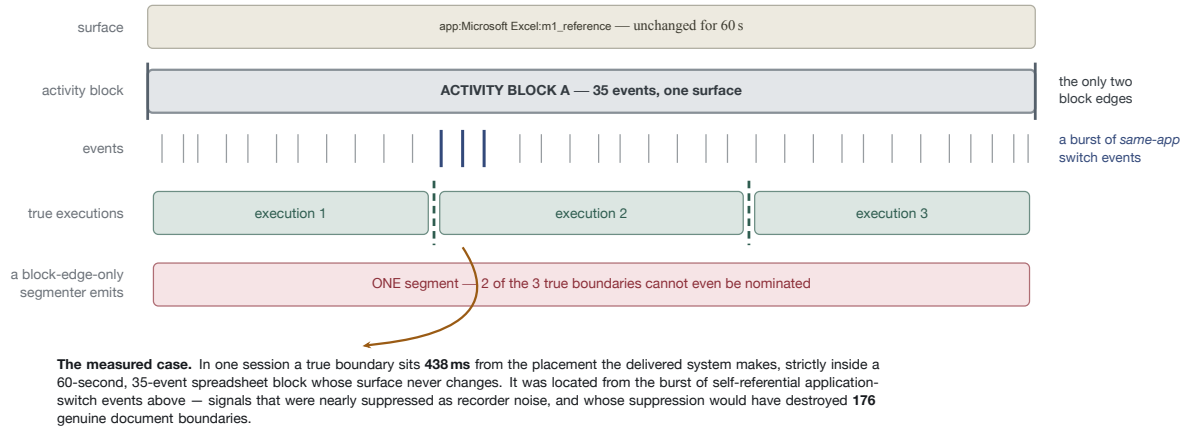


Figure 3. An activity block is not a business-process execution. The block is a container: its surface never changes, so no block-edge method can nominate either of the two internal boundaries. This single finding redirected the whole of Layer 1 — from rebuilding the representation, which measurement had exonerated, to enlarging the candidate space, which measurement had indicted.

4.3 Baselines

Six baselines were implemented *before* the method, evaluated identically, and kept in every subsequent comparison. Leave-one-session-out over all 63 annotated sessions; boundary tolerance 2 seconds.

Approach	What it does	Boundary F1	Segment-count ratio	Discovered labels
Single segment per session	one execution per session	0.000	0.04	16
Recording-chunk edges	segment where the agent flushed	0.008	0.07	32
Temporal gap threshold	segment at pauses above a fitted threshold	0.089	1.00	72
Random boundaries	matched count, random positions	0.109	1.00	56
Surface change	segment whenever the surface changes	0.247	5.03	140
Dwell-based (strongest)	segment on dwell-time structure	**0.258**	**2.83**	130
Delivered pipeline		**0.690**	**0.98**	**17**

Table 4. Six baselines and the delivered pipeline, evaluated identically. **The count-ratio column is the one that matters.** The strongest baseline achieves its F1 while emitting 2.83× as many segments as there are executions; the surface-change baseline emits 5.03×. Boundary F1 improves mechanically when a method cuts more often, so a headline F1 without a count ratio beside it is not interpretable.

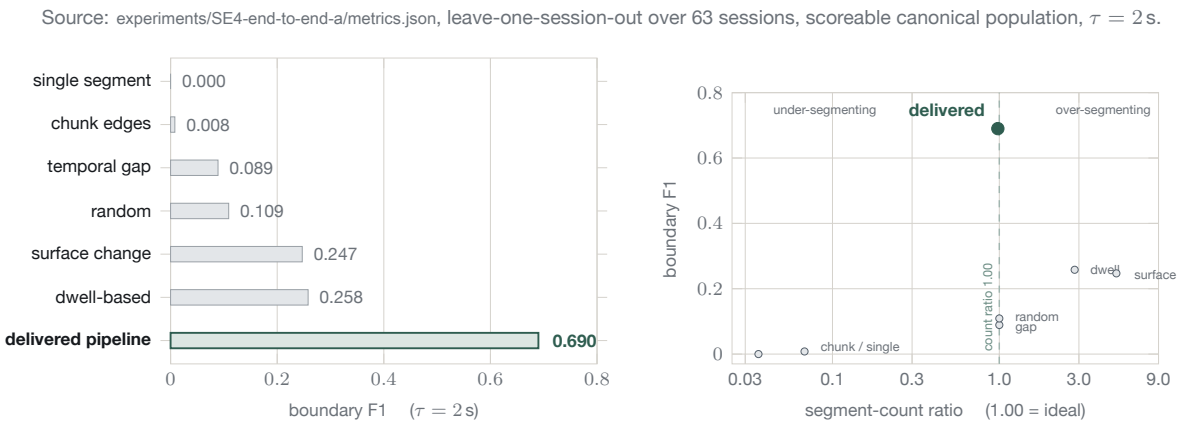


Figure 4. Boundary F1 is not interpretable on its own. Left: boundary F1 at a 2-second tolerance. Right: the same arms plotted against how often they cut. The two baselines that come closest on F1 buy it by over-segmenting by 2.8× and 5.0×; the delivered pipeline reaches 2.67× their F1 while emitting almost exactly as many segments as there are executions. A method sitting in the upper region *and* near a count ratio of 1.0 is the only one that produces a usable process inventory.

4.4 Boundary analysis

Boundary evidence was decomposed and each class measured for how much of the annotated boundary set it can reach.

- **Block edges** — surface, application or window changes. Necessary, and insufficient: 59.9% reachability on their own.
- **Intra-block event evidence** — form submissions, field resets, navigation within a single-page application, clipboard operations, keyboard shortcuts, and the self-referential application-switch bursts described above. This is what lifts reachability to **88.0%** against a measured ceiling of **90.0%** — 97.8% of what is achievable — at a candidate density of 0.165 candidates per event on the annotated data and 0.127 on production.
- **Temporal gaps** — retained as a weak feature and a baseline, never as a mechanism.

Reachability is not an accuracy number

Reachability is reported separately from segmentation quality throughout. It separates *"the candidate generator cannot propose this position"* from *"the scorer chose wrongly"*, which tells you which stage to work on. It also prevents a real error: recall is reported as **0.702 of an achievable**

0.900, i.e. 78% of what is reachable, never as 0.702 out of 1.000, which would overstate the remaining headroom by roughly a factor of three.

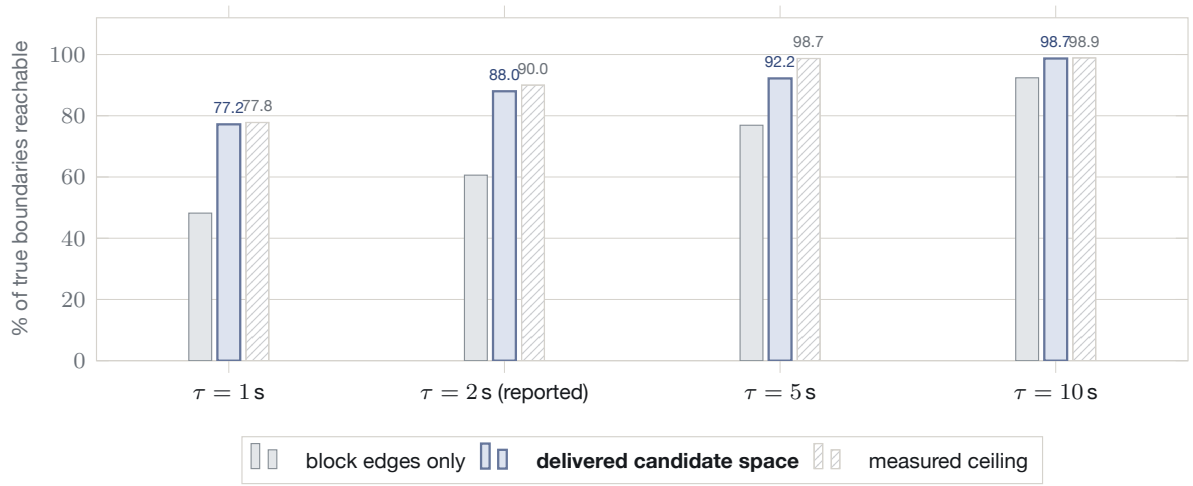


Figure 5. The candidate space, not the representation, was the ceiling. Reachability of the block-edge candidate space, of the delivered candidate space, and of the measured ceiling, at four tolerances. At the reported 2-second tolerance the event-anchored layer closes 27.4 points of the 30.4-point gap — **97.8% of what is achievable**. The delivered system's *recall* (0.702) is plotted against that ceiling rather than against 1.0.

4.5 Error taxonomy

Missed boundaries were attributed to explicit categories, at event and field granularity.

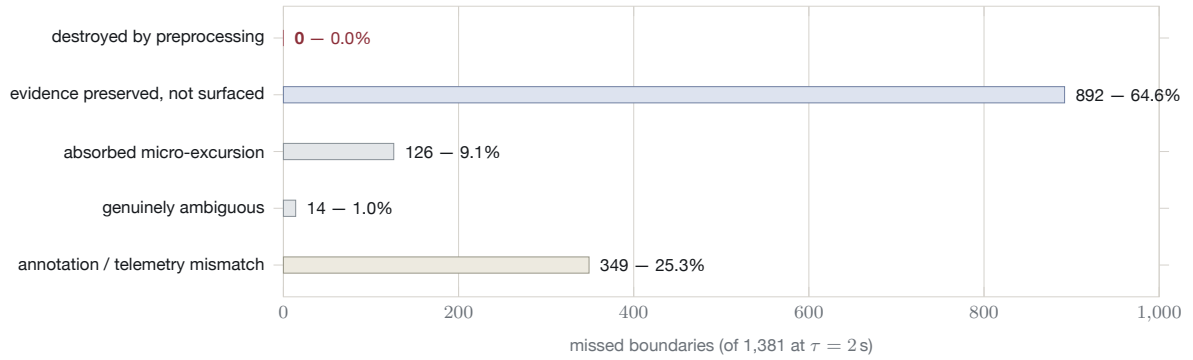


Figure 6. Where the missed boundaries actually went. "Destroyed by preprocessing" measures **zero** at both event and field level, which exonerated normalisation and redirected the entire layer. The dominant category — evidence preserved but not surfaced — is exactly what the intra-block candidate layer was built to reach. The 25.3% attributed to annotation/telemetry mismatch is a property of the reference data (annotation lead, telemetry gaps, one truncated session), not of the method, and is reported rather than excluded.

Category	Finding	Engineering consequence
Destroyed by preprocessing	**zero**, at both event and field level	normalisation was exonerated; the problem was the candidate space, not the representation. This single result redirected the whole of Layer 1
Evidence preserved but not surfaced	the dominant category (892)	motivated the intra-block candidate layer — the change that took reachability from 59.9% to 88.0%
Absorbed micro-excursions	126 short departures and returns folded into a neighbouring unit	accepted; the cost of splitting them exceeded the benefit
Genuinely ambiguous	∼10% of boundaries unreachable at a 2-second tolerance even with the full candidate space	became the measured ceiling. Chasing it would mean fitting the annotation offset

Table 5. Boundary-miss attribution and what each category caused. Categories are assigned per boundary and proven rather than assumed.

4.6 Candidate generation

Candidates are proposed from two anchor types: **event-anchored**, from the intra-block evidence classes above, and **gap-anchored**, from temporal structure. Each candidate carries the evidence that proposed it, which is what makes the later attribution and the provenance manifest possible.

Density is a reported parameter, not an accident: 0.165 candidates per event on the annotated data and 0.127 on production. A denser space raises reachability and makes the scorer's job harder; the operating point was chosen against measurement rather than by default.

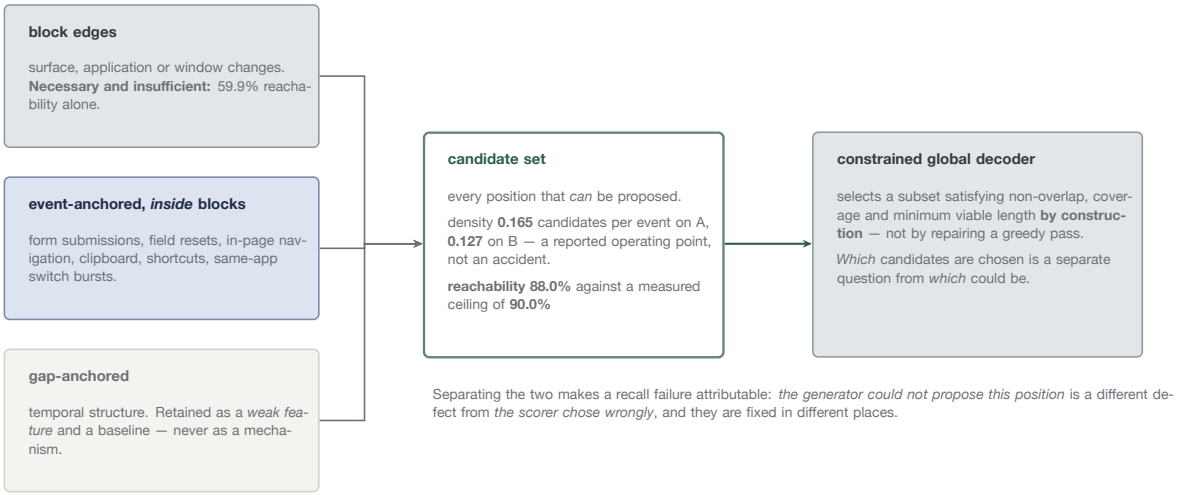


Figure 7. Candidate generation, and why it is a separate stage. Three evidence classes feed one candidate set whose reachability is measured independently; a constrained decoder then chooses a subset that satisfies the structural invariants of a valid segmentation by construction. Keeping the two apart is what makes a recall failure attributable to the stage that caused it.

4.7 Segmentation

Candidates are scored with a fitted model over positive, **negative** and composition evidence — negative evidence matters because much of the signal is "this looks like a boundary but the case continues". A **constrained decoder** then selects a subset satisfying the structural invariants of a valid segmentation (non-overlap, coverage, minimum viable length) *by construction*, rather than post-hoc repair of a greedy pass.

An **ambiguity gate** routes the residual rather than guessing at it. The gate's budget is a parameter; in the shipped configuration it consumes 3.4% of transitions.

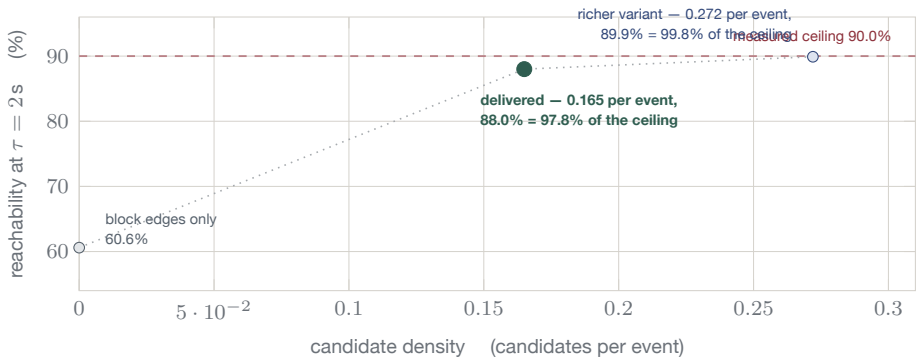


Figure 8. The candidate-space trade-off, as it was actually navigated. Adding an event-anchored layer raises reachability and candidate density together; a denser space makes the scorer's job harder. Two measured operating points are shown against the ceiling. The delivered configuration takes **97.8%** of the achievable reachability at **0.165** candidates per event; a richer variant reaches 99.8% of the ceiling but at **0.272** per event — 65% more candidates for 2 further points, which was not taken.

Cross-dataset transfer was handled as a **rule, not a value**: the operating point is calibrated structurally, so what carries from the annotated data to production is the procedure for choosing a threshold, not a number fitted to one department's behaviour. Each run writes a manifest recording the identity method used, the deliverable's content hash, and the resolved dependency versions.

4.8 Process identity

The second half of the deliverable requires that the same process receive the same label across sessions — and production has no labels to learn from. Identity is therefore **inferred**, which is where the engagement becomes a machine-learning problem rather than a rules problem: the target is a partition of unlabelled executions, the number of classes is unknown, and the only supervision available is on a different dataset with different departments, processes and applications.

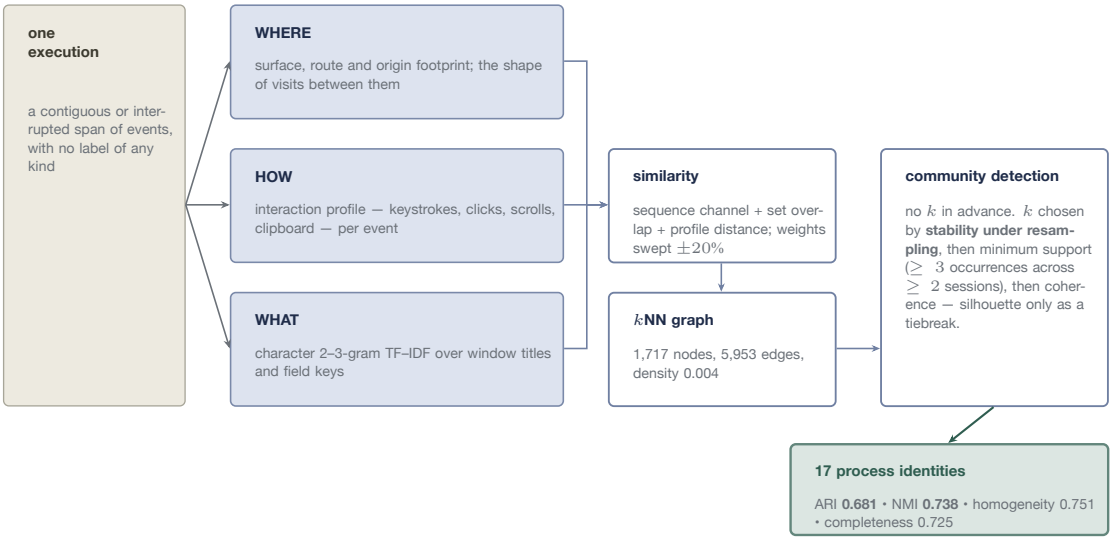


Figure 9. Process identity is the unsupervised core of Layer 1. Each execution is encoded along three named views, combined into a similarity, and partitioned by community detection over a k-nearest-neighbour graph. The number of processes is never supplied: it is chosen by stability under resampling, then minimum support, with silhouette used only as a tiebreak. Character n-grams are used for the what view because Japanese is not whitespace-delimited — avoiding a morphological analyser, and therefore a new dependency.

View	Content	Why
WHERE	surface, route and origin footprint; the shape of visits between them	two executions of one process traverse the same systems in a similar order
HOW	interaction profile — keystrokes, clicks, scrolls, navigations, clipboard, form inputs — normalised per event	the *manner* of work differs between a data-entry process and a verification process even on the same screen
WHAT	character 2–3-gram TF–IDF over window titles and field keys	character n-grams rather than words because Japanese is not whitespace-delimited; no morphological analyser, and therefore no new dependency, was needed

Table 6. The three-view execution representation. Similarity combines a sequence channel over surface shapes with set-overlap channels and a profile distance; the weights are reported and swept $\pm 20\%$ rather than presented as discovered constants.

4.9 Alternatives considered

Alternative	What it offered	Why it was not used
Gap-only segmentation	trivial, no training	1.00 count ratio but very low boundary agreement; batching defeats it
Surface-change segmentation	no parameters	over-segments at $5.03\times$ — every intra-case system visit becomes a boundary
Dwell-based segmentation	strongest baseline	$2.83\times$ count ratio; produces 130 labels, which is not a usable process inventory
Recording-chunk edges	free	measures the recorder, not the work; 0.07 count ratio
Flat k-means over executions	simple, fast	requires k in advance; the production class count is unknown
Hierarchical clustering	no k needed upfront	the cut height is the same unknown in a different coordinate
Density-based clustering	handles noise and arbitrary shapes	at a tuned radius it posts interruption-stitching 0.857 with **ARI** -0.001 — a degenerate partition that looks good on one metric
kNN graph + community detection	no k; exploits local structure; robust to varying cluster density	**chosen**: ARI 0.583 $\rightarrow$ 0.681 with the tie-break, at 17 labels
Language model for identity	semantic understanding of screens	sized and declined three times on measured ceilings — § 5.3

Table 7. Alternatives for segmentation and identity, with the measurement that decided each. Oracle-k results on the annotated data are reported separately from unknown-k results on production, because conflating them inflates the apparent result.

5 Layer 1 — results, and what they cost to believe

5.1 Boundary placement

Leave-one-session-out over all 63 annotated sessions, scoreable canonical population.

Tolerance	Precision	Recall	F1	Reachability ceiling	Recall / ceiling
1 s	0.534	0.551	0.543	77.8%	70.9%
2 s	**0.680**	**0.702**	**0.690**	**90.0%**	**77.9%**
5 s	0.729	0.752	0.740	98.7%	76.2%
10 s	0.781	0.806	0.794	98.9%	81.5%

Table 8. Boundary placement across the full tolerance curve, each row paired with the reachability ceiling at the same tolerance. Results are reported at 2 s because the annotation leads the nearest retained event by ≈176 ms at the median — below 1 s the metric measures the annotation offset rather than the method.

Source: experiments/SE4-end-to-end-a/metrics.json. Segmental F1 at IoU 0.5 **0.768** • WindowDiff **0.470** • Pk **0.257** • per-session F1 0.683 ± 0.115 • candidate-ranking AUC 0.979.

Metric meanings, since these are not interchangeable. *Boundary F1* scores whether a cut was placed within the tolerance of a true cut — it is about *positions*. *Segmental F1 at IoU 0.5* scores whether a produced segment overlaps a true segment by at least half — it is about *whole units*, and is stricter. *WindowDiff* and *Pk* are sliding-window disagreement rates from the text-segmentation literature where lower is better; they penalise near-misses less harshly than exact-match metrics and are included because boundary F1 alone is sensitive to tolerance choice.

5.2 Process identity

On the matched population (match rate 0.760, boundary F1 0.690 — unchanged across all three arms, so the comparison isolates identity).

Approach	ARI	NMI	Homogeneity	Completeness	Labels
Structural-signature labeller	0.431	0.563	0.622	0.514	51
kNN graph + community detection	0.583	0.688	0.679	0.698	**15**
+ saturation tie-break (delivered)	**0.681**	**0.738**	**0.751**	**0.725**	17

Table 9. Process identity, three arms on an identical segmentation. **Homogeneity and completeness both rise at every step**, which is the check that the improvement is not over-merging — ARI alone cannot distinguish "better clusters" from "fewer, larger clusters", and reporting it alone is how clustering results get oversold.

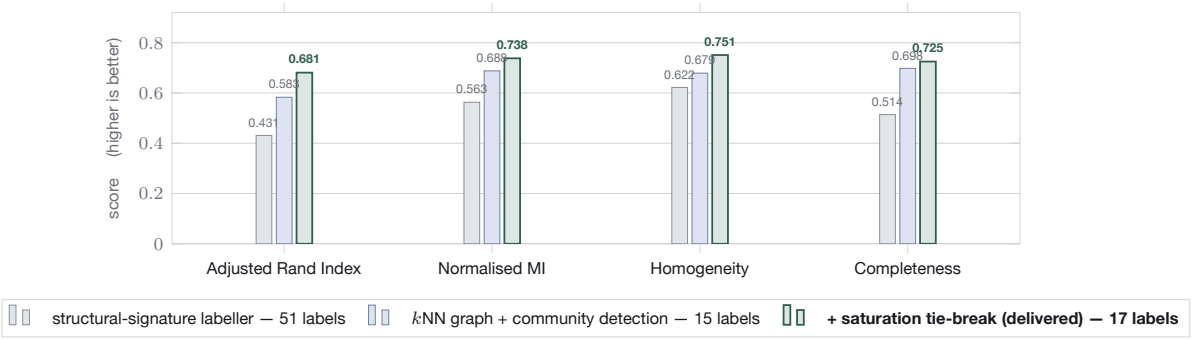


Figure 10. Four metrics, not one. Reporting ARI alone cannot distinguish a better partition from a coarser one. Homogeneity (does each discovered cluster contain only one true process?) and completeness (does each true process land in only one cluster?) rise together at every step, and the label count falls from 51 to 17 — which is what makes the result a usable process inventory rather than a better number.

The baseline that appears to win, and does not. The dwell baseline posts matched-segment **ARI 0.797** against the pipeline's 0.681. Rather than dropping the comparison we pursued it. Matched-segment ARI is computed only over segments that matched a true segment, so a method that segments

rarely is scored on a favourable subset. The reductio is in the table: the recording-chunk baseline posts matched **ARI 1.000 at a match rate of 0.0017**.

Arm	Matched ARI	Match rate	Event ARI	Time ARI	Count ratio	Labels
Delivered pipeline	0.681	0.760	**0.297**	---	**0.98**	17
Pipeline before tie-break	0.585	0.760	0.267	0.269	0.98	15
Structural labeller	0.431	0.760	0.182	0.184	0.98	51
Dwell baseline	**0.797**	0.597	0.250	0.200	2.83	130
Surface-change baseline	0.770	0.562	0.244	0.162	5.03	140
Gap baseline	0.682	0.134	0.119	0.103	1.00	72
Random baseline	0.362	0.320	0.121	0.080	1.00	56
Chunk-edge baseline	**1.000**	**0.0017**	0.005	−0.006	0.07	32
Single-segment baseline	---	---	0.003	−0.008	0.04	16

Table 10. Why matched-subset identity metrics are not comparable across arms with different match rates. **Event ARI** — computed over all events rather than a matched subset — is the comparable metric, and it puts the pipeline ahead at 0.297 against 0.250. Two honest qualifications: the identity margin (1.19×) is far narrower than the 2.67× on boundary placement, and the dwell baseline reaches its 0.250 with 130 labels at a 2.83 count ratio, which is not a process inventory anyone can act on.

5.3 Where a language model was sized, and why it was declined

This is the most important methodological point in Layer 1, and it is not a preference.

Proposed role	Measured ceiling	Verdict
Merge and rename identity labels	perfect-oracle merge reaches **ARI 0.478**	**below** the 0.583 the clustering already achieves — bounded before a line was written
Adjudicate ambiguous boundaries	perfect oracle worth +0.026 **F1** at the full 15% gate budget; +0.004 at the 3.4% actually used	perfect adjudication of the *entire* budget is worth 2.6 points
Supply missing semantics for identity	a linear probe over exactly the clusterer's features recovers the reference process code at **0.891 accuracy / ARI 0.786**	the information is **present, not missing**

Table 11. Three proposed roles for a language model in the inference layer, and the measurement that closed each. The third is decisive: if a linear model over the same features recovers the process code at 0.891, the residual is an *unsupervised-objective* gap, not a semantics gap a language model could fill.

> **The scoped next step, deliberately not built**
>
> **Metric learning from bounded model-generated pair labels** — use a language model to label a sample of execution *pairs* as same-process or different-process, fit the view weighting to that signal, and ablate against the current unsupervised rule. That places a model exactly where the measurement says the gap is. It was recorded rather than half-built.

5.4 The residual, named as a finding

One experiment set out to fix the largest identity defect and instead falsified the mechanism behind it. The worst cross-process pairs have a surface Jaccard similarity of **exactly 1.00** — identical surface sets, a mean of 1.35 shared surfaces, and that is every surface either execution has. No term re-weighting can separate identical sets, and inverse-document-frequency weighting measurably made things worse (separability Cohen's d 1.535 → 1.508; ARI −0.031). It is implemented, tested, ablatable, and **off**.

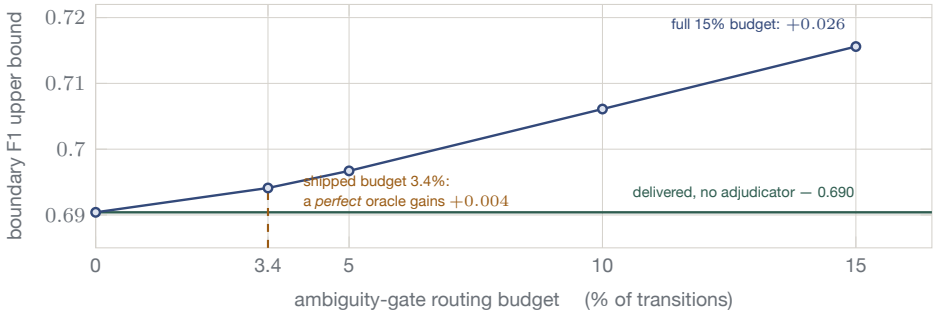


Figure 11. What perfect adjudication would be worth. The upper bound on boundary F1 if an oracle resolved every transition routed to the ambiguity gate, plotted against the gate's routing budget. At the shipped budget of 3.4% a *perfect* adjudicator gains $+0.004$ F1; at the maximum 15% budget it gains $+0.026$. A real language model would achieve some fraction of that. The decision to ship without one is a reading of this curve, not a preference.

The tie-break that does ship attacks the same pairs from the other side: where the where view is saturated it carries no information, so it is discarded and the pair is decided on what and sequence. That bought $+0.096$ **ARI**, the largest single identity gain in the engagement — and still leaves the top-similarity cross-process pairs co-clustered.

The finding, and it is a limit of the medium rather than of the method

Executions with identical structural footprints and different business purpose are observationally equivalent at the UI-event level. Two operators opening the same page and typing into the same fields for two different business reasons emit the same events. No amount of clustering removes it. It is also a direct preview of the correction in § 9.

5.5 Additional limitations of this layer

- **Interruption stitching 0.337** (improved from 0.122), at count ratio 0.98 and 17 labels. There is no resumption model: fragments of one suspended execution share a label only by resemblance. The metric must be read with the label count, because coarser labels raise it mechanically — which is exactly why density-based clustering posts 0.857 with ARI -0.001.
- **Residual decomposition.** Of 411 misgrouped matched executions, roughly **145 are probe-hard** (the supervised probe misses them too — genuinely ambiguous) and roughly **266 are clustering-specific headroom** a better unsupervised objective could still reach.
- **Scaffolding retained deliberately.** 98 of 1,717 executions (5.7%) carry machine setup, launch or theme signatures; 44 are in the matched population and account for 9.5% of residual error — a **2.9× enrichment**. They stay, because excluding inconvenient executions is fitting the evaluation set.

5.6 An operational failure worth recording

The graph identity method's numerical dependencies were nominally optional. On a host where the virtual environment failed to resolve, the pipeline fell through to an ambient interpreter, the labeller caught the resulting import error, and **a different deliverable shipped with exit code 0** — identical segment spans, 33 labels instead of 13. The fix was to make those dependencies hard requirements and have the labeller *refuse to run* without them, plus the run manifest recording the method, deliverable hash and resolved versions.

A silent fallback is worse than a crash

This is the same principle the delivered product applies to selector drift and to unresolvable documents. After this, Layer 1 was **frozen** and never modified again.

5.7 Key findings, and why they mattered downstream

Finding	Evidence	Consequence for later layers
Recording boundaries are not business boundaries	chunk-edge baseline: count ratio 0.07, event ARI 0.005	chunks dissolved at normalisation; nothing downstream can learn them
Activity blocks are useful but insufficient	block-edge reachability 59.9%	boundary inference moved below the block; blocks retained as context
Execution boundaries occur *inside* blocks	a true boundary 438 ms from placement, inside a 60 s single-surface block	the intra-block candidate layer — the change that made the result possible
Telemetry preserves more than naive abstractions expose	"destroyed by preprocessing" measured at **zero**	the ceiling was in the candidate space, not the representation. Redirected the entire layer
Process identity is a similarity and discovery problem	production has no labels; annotated codes are non-transferable	unsupervised graph clustering, with k chosen by resampling stability
Identical footprints with different purpose are indistinguishable at event level	surface Jaccard exactly 1.00 on the worst pairs	**predicted the Layer-2 correction**: if two processes can look identical in telemetry, telemetry similarity cannot be trusted as business equivalence without external validation
A language model was not the missing piece *here*	three measured ceilings: 0.478, +0.026 F1, probe 0.891	the budget went into representation and objective instead — and the model was placed later, where measurement said it belonged

Table 12. Layer-1 findings and their downstream consequences. The sixth row is the bridge: Layer 1 had already established, on the annotated data, that UI-level evidence can be identical for different business purposes. Layer 2 then walked into exactly that trap on the production data — and recognised it because the limit had already been named.

6 Layer 2 — business-process analysis of the production dataset

The frozen segmentation and labelling were applied to Dataset B, and the resulting executions aggregated into a business inventory: **37 measured columns per module**, covering frequency, aggregate and per-disposition effort, operator spread, variant structure, interaction volumes, cross-module contamination, and session concentration.

6.1 The module inventory

Twelve modules across three business systems. The principal candidates, with values as measured.

Module	System	Exec.	Sess.	Oper.	Share of meas. time	Disp.	Sec. per disp.	Purity
Contract management	Logistics	26	8	4	**15.19%**	65	17.9	0.85
Expense settlement + payroll change	HR	20	10	4	14.33%	**87**	**12.6**	0.82
Invoice approval	Finance	25	8	4	12.25%	52	18.0	0.88
Onboarding verification	HR	23	8	4	9.32%	42	17.0	**0.90**
Purchase-order management	Finance	13	8	4	8.18%	52	12.0	0.69
Inventory management	Logistics	13	6	4	7.55%	40	14.4	0.76

Table 13. The six principal modules of the twelve-module inventory. *Module purity* is the share of an execution's business activity belonging to its dominant module — the measure of how cleanly the segmentation isolated it. Time shares are **comparative only**: waiting times were compressed in the capture environment, so no absolute duration claim is possible.

Source: docs/STEP2_PROCESS_INVENTORY.csv (37 columns per module; the full inventory carries all twelve).

Three denominators, kept strictly apart

An **execution** is one pass through a procedure; a **disposition** is one case dispositioned; an **event** is one telemetry record. The ratio between the first two ranges from **1.83 to 4.35** across modules, so a claim quoted in the wrong denominator can be wrong by a factor of two.



Figure 12. The production process inventory, and the tension in it. Each module is placed by its share of measured time (the value axis) against its module purity — the repeatability signal available from telemetry — with marker area proportional to dispositions. The upper-right quadrant, where a high-volume process is also a clean, repeatable one, is empty. That absence is the finding, and it is why the framework in § 8 reports two axes and refuses to average them.

6.2 The tension the analysis surfaced

The largest process by time is the least automatable. The resident-tax family accounts for **19.2% of segmented time** and is the only high-impact label, and it bands **lowest** on suitability: route Jaccard 0.35, surface Jaccard 0.13 — the least repeatable structure in the dataset.

No process reaches high on both axes

This is exactly the frequency-versus-automatability tension the literature describes, appearing intact in the client's data, and it is why the analysis reports two axes and no composite score. Collapsing them into one number would have hidden the finding that matters most for prioritisation.

6.3 What the analysis could not establish

Three columns of the inventory are filled with UNAVAILABLE, deliberately: **business volume**, **monetary value** and **error rate**. None is recoverable from telemetry. Leaving them visibly empty rather than estimated is what keeps the rest of the table usable.

Evidence	Directly observable?	Known failure mode
Event type, timestamp, sequence	**yes**	agent gaps; one input event type is documented as unreliable
Active application / process name	**yes**	an application is not a process
Window title	**yes**	a filename is not a business meaning; the title lags the tab at switch in 18–23% of blocks on an origin
Browser route (/payroll-items etc.)	**yes**	**the central trap**: one route renders three different business functions
Element id (btn-pi-ok, pi-note)	**yes**	only 12 distinct ids, route-derived, so they inherit the route ambiguity
Form field *value*	**no** — null on all 608 inputs	no entered value is recoverable anywhere
Clipboard *content*	**no** — null on all 872 events	only text length survives (24–80 chars, median 42)
Screen headers, table cells, badges, buttons	screenshots and extracted screen text only	not machine-available at scale; 4.59% text coverage
Business volume, monetary value, error rate, FTE cost	**no**	any figure would be invented — recorded as UNAVAILABLE

Table 14. The evidence matrix, abridged. Each row records what the signal is, whether it is directly observable, and the failure mode that stops it being trusted as business semantics. The route row is the one that later cost a full re-analysis.

Source: docs/STEP2_EVIDENCE_MATRIX.csv.

7 Literature and industry review

The framework in § 8 was not invented. It was assembled from research on process and task mining, robotic-process-automation candidate selection, and — because this is a Japanese client — Japanese public-sector and vendor guidance. Full citations in § 25.

7.1 Process and task mining from interaction logs

The foundational survey of **robotic process mining** [1] frames a seven-stage pipeline: recording, noise filtering, **segmentation**, simplification, candidate routine identification, executable routine discovery, and compilation. Its most useful contribution to this engagement is that it names **segmentation as an open problem**, precisely because user-interaction logs lack case identifiers. Process mining proper assumes a case id in every event; task mining as sold by vendors assumes a subject-matter expert has already nominated the task. This engagement had neither, which is why Layer 1 exists and why it consumed a third of the budget.

Discovering automatable routines from user-interaction logs [2] supplies the formal condition this engagement tested: a routine is automatable when its parameters are either constant or derivable from earlier observed values. That condition is a statement about *data values* — and Dataset B's values are redacted, which forced the substitution described in § 7.6.

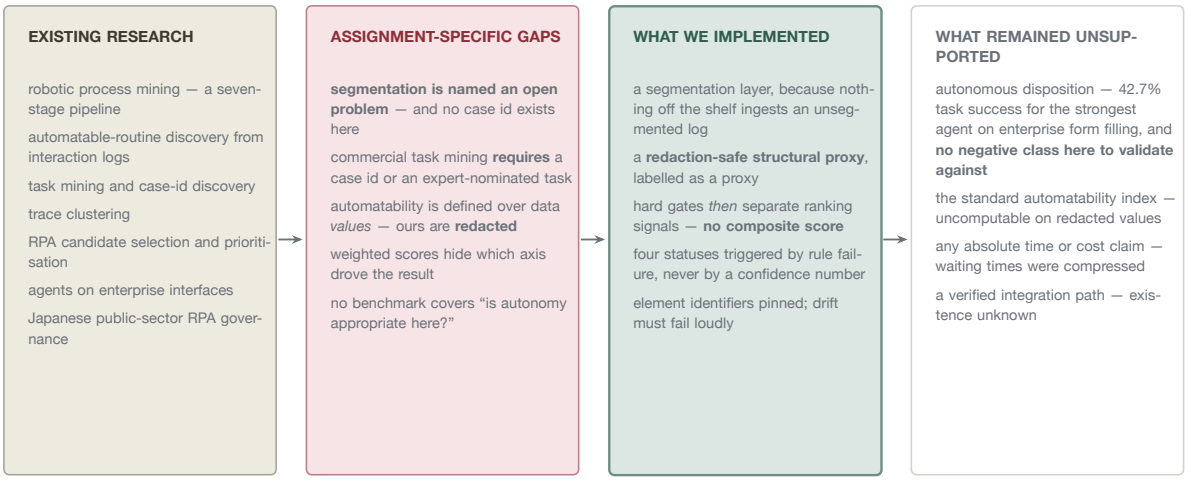


Figure 13. How the literature informed the engineering. Reading left to right: what the research provides, where it stops short of this problem, what was built in response, and what the evidence never licensed. The purpose of a literature section in an engineering report is the third and fourth columns — every element in them traces to a specific finding in the second.

Vendor documentation confirmed the constraint rather than contradicting it: UiPath's task-mining analysis guide [3] and Power Automate's process-mining requirements both mandate a case identifier column. Nothing off the shelf ingests an unsegmented interaction log.

7.2 What makes automation candidates succeed

The recurring characteristics across the literature are consistent and unsurprising: high volume, high repeatability, structured and digital input, stable applications, low exception rates, low requirement for judgement, and prior process standardisation. UiPath's Automation Hub idea score [7] operationalises a commercial version of this, combining expected benefit with implementation complexity.

> **The more useful finding is the negative one**
>
> The maintainability study [9] identifies **change in the automated applications' user interfaces as the dominant driver of long-run maintenance cost** — not logic complexity. That single result is why the delivered system treats interface drift as something that must fail loudly, and why an "adaptive recovery" feature was explicitly rejected (§ 17).

The systematic review of RPA in business process management [8] and the contemporary-themes survey [10] both emphasise that automating a poorly-understood process encodes its defects.

7.3 Prioritisation methodology

Family	Example	What it offers / why not adopted wholesale
Weighted scoring	Wanner et al., *Process Selection in RPA Projects* [4]	transparent, auditable, easy to communicate • *but* weights are the analyst's judgement wearing arithmetic; a single score hides which axis drove it
Multi-criteria decision analysis	AHP + TOPSIS applied to automation selection [6]	principled aggregation of incommensurable criteria • *but* requires pairwise-preference elicitation from stakeholders we did not have access to
Quantified candidate mining	*Digging for Gold in RPA Projects* [5]	derives candidate quality from the log itself • *but* assumes logged data that includes the values; ours are redacted

Table 15. Three prioritisation families examined. The method adopted takes the criteria from all three and **refuses the aggregation step**; § 8 explains why.

7.4 Autonomous agents on enterprise interfaces

WorkArena [13] benchmarks language-model agents on 33 enterprise knowledge-work tasks — including form filling and service-catalogue ordering — on a real platform. The reported figures are **42.7% task success for GPT-4 with an optimised configuration and 17.9% for the open-source state of the art**, with failure modes including enterprise HTML exceeding 100,000 tokens and agents struggling to manipulate the interface.

> **The single most decision-relevant external number in the engagement**
>
> Against a deterministic path driving element identifiers that the client's own telemetry already contains, an agent on the execution path is worse on success rate, reproducibility, explainability and auditability — and its failures are *silent* rather than visible. For a process with statutory exposure, that is disqualifying, and it is disqualifying on a measurement rather than on a preference.

AWS's intelligent-document-processing reference architecture [14] contributed a second design constraint: route to human review when **business rules fail**, treating model confidence as one condition category among several and only where extraction is probabilistic. Our document-to-record mapping is deterministic, so a confidence number attached to it would be invented precision. This is the origin of the four-status vocabulary in § 14.

7.5 The Japanese enterprise and public-sector context

The client is Japanese and the context changed the analysis in three concrete ways.

The problem has a name, and the name carries a solution catalogue. Japanese enterprise IT calls duplicate data entry across core systems 二重入力 (*nijū nyūryōku*, duplicate entry) and its documented consequence 転記ミス (*tenki misu*, transcription error), occurring between 基幹システム (*kikan shisutemu*, core systems) [20]. Framing the observed work as "document-to-form transcription" would have imported the wrong catalogue — intelligent document processing and optical character recognition, which solve a problem we do not have, since the source is a structured list and not a scanned document — and would have hidden the multi-system dimension that is the actual source of waste.

Integration may already exist, and that would change the recommendation. Japanese vendor material documents CSV import between payroll and accounting packages as a *normal* capability [21]. Nothing in the telemetry indicates whether the client's systems expose one: observing that staff use the interface is not evidence that no other interface exists. The uncomfortable case is stated deliberately — if a bulk import already exists and staff are nevertheless hand-keying, the finding is a **process** finding, not a technology one, and the correct recommendation is partly training and process change. This is why the delivered architecture makes the execution backend an interface rather than hard-wiring one (§ 13.4).

Governance guidance is explicit about the failure mode. The Ministry of Internal Affairs and Communications' RPA introduction guidebook for local government [16] leads with 業務の整理・見直し (*gyōmu no seiri · minaoshi*) — organise and review the work *before* automating it. Mitsubishi Research Institute's widely-cited analysis [17] names the outcome of ignoring that advice: 野良ロボット (*nora robotto*, "stray robots") — undocumented automations nobody owns. Japanese practice also insists that verification survives automation: even with system integration, confirmation of the posting result remains necessary. Human review is not optional garnish in this context.

Labour-market material [22, 23, 24] supplies the macroeconomic motivation — demographic decline making back-office capacity structurally scarce — and is cited as context, not as a source of any number used in this report.

7.6 Synthesis

Research finding	Implication for this engagement
Segmentation of interaction logs is an open problem; case ids are absent	Layer 1 is unavoidable and deserves a third of the budget
Commercial task mining requires a case id or an expert-nominated task	nothing off the shelf applies; the pipeline had to be built
Automatability is defined over data *values*	values are redacted → the standard index was replaced by a **redaction-safe structural proxy**, and labelled as a proxy
Interface change dominates long-run maintenance cost	drift must fail loudly; adaptive recovery rejected; element identifiers pinned to a manifest
Weighted scores hide which axis drove the result	hard gates plus separate ranking signals; **no composite score**
Human review should be triggered by rule failure, not by confidence	four statuses, no confidence score anywhere in the product
Agents reach 42.7% on enterprise form filling	no model on the execution path; the irreversible action stays human
The Japanese framing is duplicate entry across core systems, not transcription	the multi-system dimension stays visible; document-processing solutions rejected as ill-fitting
CSV or API integration is a common capability and is unverified here	the execution backend is an interface; a process finding remains a possible answer
Public guidance: organise the work before automating; beware unowned robots	governance, provenance and audit are first-class, not added later
Verification survives automation	the product automates *preparation*; the confirmation remains human by design

Table 16. Every row is a research finding paired with the specific engineering decision it produced. This table, not the citation list, is what the literature review was for.

8 The automation opportunity framework

8.1 Why not a single score

A weighted composite would have produced a ranking, and the ranking would have concealed the most important thing the data says: **the largest process by time is the least repeatable**, and no process scores well on both axes. Averaging those two facts produces a mid-ranked candidate and destroys the finding. The framework therefore has three distinct parts.



Figure 14. Hard gates, ranking signals and open business questions — in that order, and never averaged. Gates are binary and applied first, so an attractive candidate with insufficient evidence is excluded rather than discounted. Ranking signals stay on separate axes. The questions telemetry cannot answer travel with the candidate instead of being assumed away; one of them blocked a candidate outright.

8.2 Hard gates — evidence sufficiency, applied before ranking

Gate	Requirement	Rationale
Multi-session, multi-operator	observed in more than one session and by more than one operator	a single operator on a single day is one person's habit, not a process
Identity confidence	module purity above threshold	a contaminated label makes every derived figure a blend of different work
Interaction substance	the executions are real work, not navigation	otherwise dashboards and menu traversal rank as processes
Reference recoverability	the governing document or reference is identifiable	without it, neither validation nor the product is possible
Deciding-input observability	the variable a rule needs is present on the record	**this gate is what § 10 turned into a measurement**

Table 17. The five hard gates. Two modules were excluded on the first gate; one — purchase-order management — failed identity confidence at purity 0.69, and its exclusion is itself informative, because the contamination came from the five-way over-merge described in § 9.

8.3 Ranking signals, kept separate

Criterion	Evidence source	Observability	Role
Frequency	executions and dispositions per module	direct	value axis
Aggregate effort	share of measured time	direct, **comparative only**	value axis
Operator spread	distinct operators, session concentration	direct	value axis and risk
Repeatability	module purity, route and surface Jaccard	direct	feasibility axis
Variance	interquartile range over median duration	direct	feasibility axis and variant signal
Structuredness	queue tables, typed fields, published reference lists	direct from screen text	feasibility axis
Target stability	recovered element identifiers, single vs multiple origins	direct	feasibility axis and maintenance risk
Cross-system transitions	distinct applications per execution	direct	value axis and complexity
Human judgement	disposition vocabulary; whether the deciding input is visible	**partly unobservable**	safety axis
Exception profile	badges, markers, deferral behaviour	direct but sometimes uninterpretable	safety axis
Maintenance exposure	number of systems, interface volatility	inferred from literature	cost axis
Business volume, monetary value, error rate	**not recoverable**	**absent**	recorded as UNAVAILABLE

Table 18. Eleven ranking signals and three that do not exist in this data. The observability column is why the framework refuses to aggregate: two of the signals are only partly observable and three are absent entirely, so a composite would silently weight a guess against a measurement.

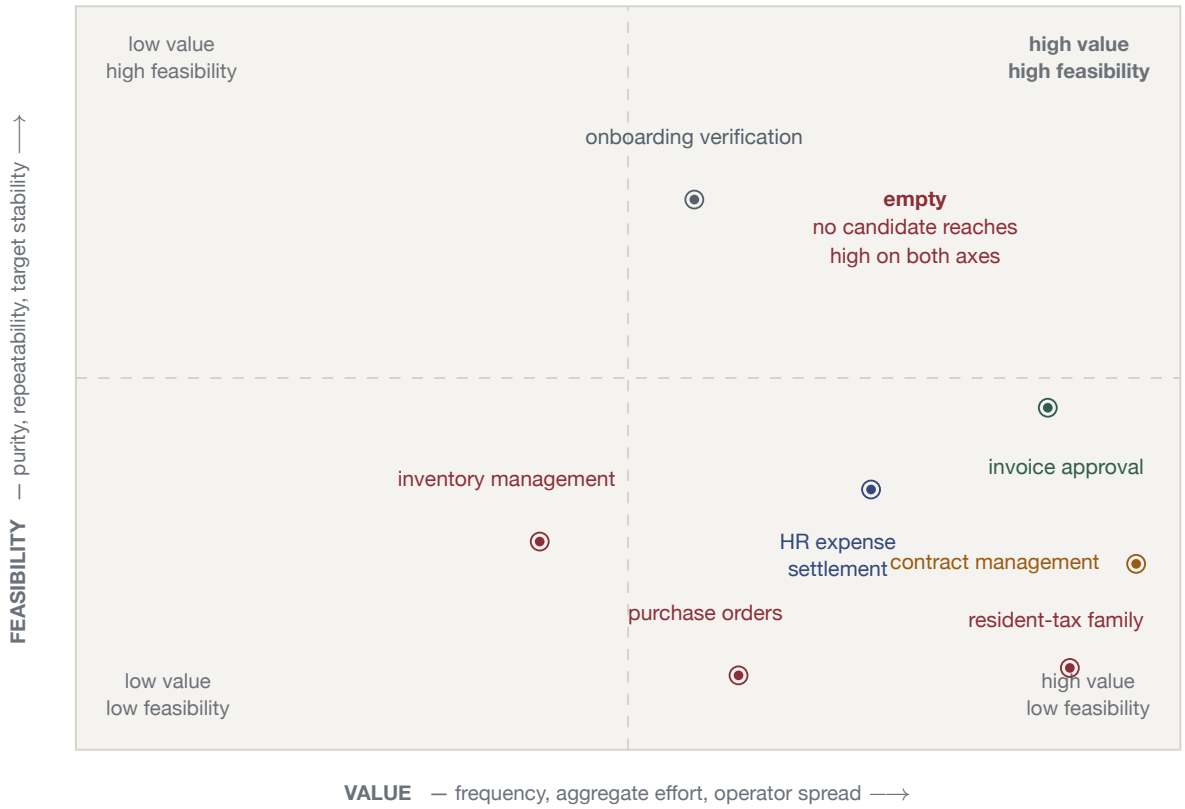


Figure 15. Candidates placed by assessment, not by score. There is no composite number behind any position here — placement reflects the gate outcomes and the ranking signals of Tables 17 and 18, and the figure would misrepresent the method if it implied otherwise. The empty upper-right quadrant is the finding. Invoice approval was ranked first not because it is the largest — it is third by time — but because it is the only candidate that passes *every* gate including the deciding-input gate.

8.4 The candidate matrix

Module	Gates	Recommended modality / AI relevance	Blocking unknown
Contract management (LO)	pass; deciding-input gate **fails**	prepare-and-present + human decision. **Low** — document selection is a lookup, the comment is a template	contract value absent from the record, so the authority rule cannot be applied
Invoice approval (FI)	**pass, all gates**	propose → validate → human approves → deterministic execution. **High** — entity resolution, clause retrieval, decision proposal	a status badge appears to contradict the written rule; vendor names do not join exactly
Expense settlement (HR)	pass; deciding-input gate unknown	human-in-the-loop decision support. Medium	is this one process or two?
Onboarding verification (HR)	pass, partial	deterministic rules engine. **Low — AI would be decoration**	are the conditions only the two observed?
Inventory management (LO)	marginal	**none** — insufficient evidence	54% of executions come from one session
Purchase-order management (FI)	**fail** identity (0.69)	**none yet**; potentially high, currently unevaluable	**what the warning marker means — blocks the candidate entirely**

Table 19. The candidate matrix. Two entries are worth reading as judgement rather than output: onboarding verification has the *best* identity purity (0.90), the lowest variance and the cleanest two-application footprint — and is marked *AI would be decoration*, because a deterministic rules engine covers it.

Source: docs/STEP2_CANDIDATE_MATRIX.csv. The full matrix carries eighteen columns per candidate, including the validation mechanism and the recorded unknowns.

9 The correction that mattered

9.1 The candidates as first analysed

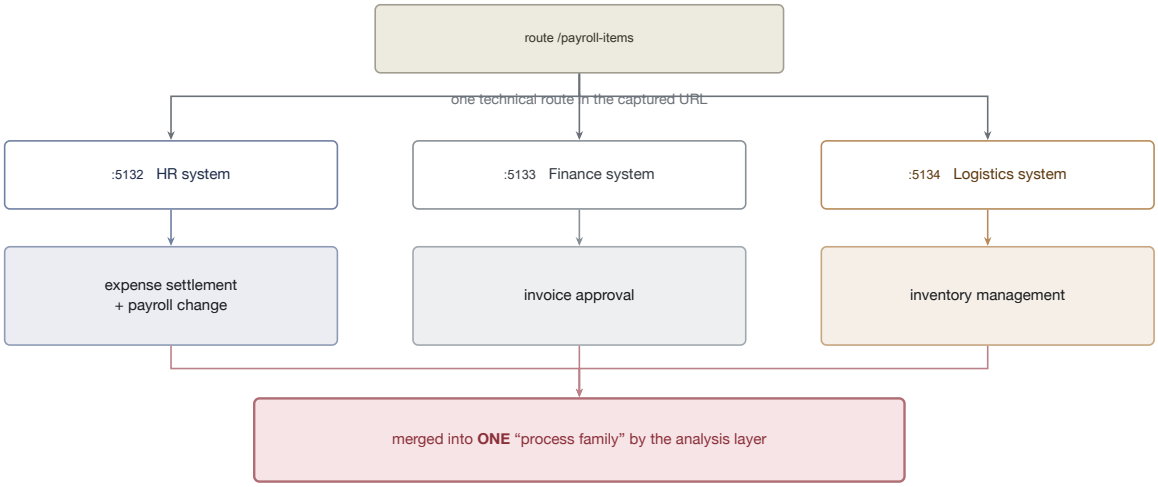
The initial analysis produced a coherent picture: a small number of high-frequency modules, four operators spread across them, templated comments, single-origin element identifiers, and a clean selector-level automation surface recoverable from the logs alone. One striking observation was that the apparent cross-system complexity was largely a browser, a word processor and operating-system shells — not a genuine multi-application tangle.

9.2 The uncomfortable observation

The most frequent process was also the least structurally repeatable, and the module with the best identity purity was the one where automation would add least. That is not a comfortable result to report, and it is the result that made the two-axis framework necessary.

9.3 A technical route is not a business process

The inventory aggregated executions by **label**, and labels were generated from the URL route. Three separate business systems serve the route /payroll-items.



The defect was one function, in the analysis layer, not the model. The aggregation key was derived by stripping a suffix from the *display name* — and the display name was origin-blind. The segmentation layer had never lost the distinction: 70% of its distance metric is origin-qualified. The analysis discarded it in one line.

Technical route similarity ≠ business-process identity

What caught it was validating the representation against the interface the human was actually looking at — screen text and screenshots. Layer 1 had already named the same limit on the annotated data (§ 5): executions with identical structural footprints and different business purpose are observationally equivalent at the UI-event level.

Figure 16. The methodological lesson of the engagement. Records from invoice approval in Finance, expense settlement in HR and inventory management in Logistics had been merged into a single “process family”. A model can be internally consistent, properly measured, and still wrong about the world — and the only thing that catches that class of error is validating the representation against the interface the human was actually looking at.

The response

The analysis was re-performed *from scratch* rather than patched, with every prior conclusion treated as a hypothesis. **42 claims were re-audited: 16 held, 11 were contradicted.** An independent review was run against primary sources. The corrected picture: two of the merged labels resolve to invoice approval (92% dominant) and HR expense settlement (77%) — two processes in two systems, not one family; another resolves to contract management (89%); onboarding verification maps at 100%; and the resident-tax family has **no dominant module at all (0.26)** — a five-way over-merge, which also explains the purity-0.69 gate failure in Table 19.

9.4 The correction was itself corrected

The key we switched to — the URL — turned out to be wrong too. Over 3,011 extracted records, three candidate module keys were compared: the URL route, the module name rendered in the screen's own breadcrumb, and the record-identifier prefix. The prefix-to-module mapping was **derived** from 211 queue captures rather than assumed, and all 13 prefixes resolve at purity 1.00.

Comparison	Comparable records	Agree	Rate
Screen header ↔ record prefix	2,904	**2,904**	**1.000**
URL ↔ record prefix	2,704	2,139	0.791
URL ↔ screen header	2,630	2,067	0.786

Table 20. Three-key triangulation over 3,011 extracted records. Only two patterns occur: all three agree (2,064 records), or **the URL dissents alone** (552 records). The screen header and the prefix never disagree with each other — and that asymmetry is the signature of one wrong key, not three partial views.

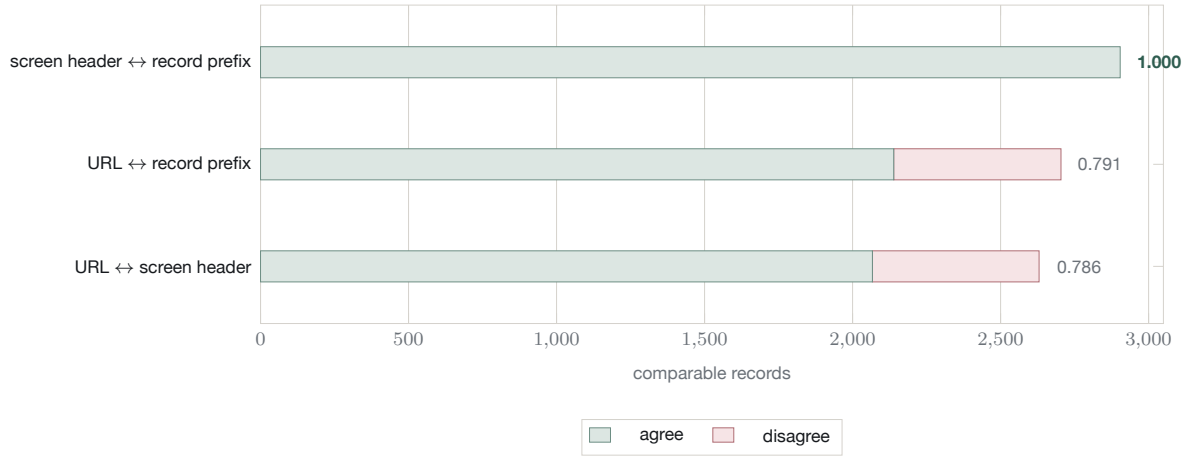


Figure 17. One wrong key, not three partial views. If the three keys were each right about a different question, disagreements would be distributed among all three. They are not: every one of the 552 disagreements is the URL dissenting alone, and the screen header and record prefix agree on 2,904 of 2,904 comparable records. The leading benign explanation — shared cross-module queues — was tested directly and refuted: of 33 captures containing more than one record prefix, **zero** cross a module boundary.

Two further consequences. First, a **twelfth module was missing from the route map entirely** — budget variance analysis is declared on screen in 11 captures and carries 132 record rows at purity 1.00, but had no map entry and therefore zero executions by construction. The omission was invisible precisely because it produced no error. Second, the two keys trade off: the URL assigns 165 of 183 executions and the prefix only 107, because extracted screen text reaches 4.59% of events.

Neither key dominates, so per-module counts are reported as ranges

The prefix is more accurate and less complete; the URL is more complete and less accurate. No single-key inventory is defensible. Contract management is between **8 and 26** executions depending on the key, and the spread is *coverage*, not measurement noise.

9.5 The hypothesis this produced

The analysis converged on a single attractive proposition, and it is stated here in its strong form because § 10 exists to test it.

The strong hypothesis

Operators apply recoverable written rules to record variables, and an AI system can learn that decision function.

If true, the product is decision automation and the engagement has a high ceiling. It was too important to assume.

10 Layer 3 — testing the hypothesis before building on it

10.1 Why this layer exists

The strong hypothesis requires four things to be true simultaneously. Each was made a **gate**, and all four were declared — along with an evidence floor of $n \geq 8$ per module, 10,000 permutations, and a fixed random seed — **before any of the numbers were inspected**. "Untestable" and "insufficient evidence" were declared first-class outcomes in advance, so that neither could be quietly converted into a weak positive.

10.2 The evidence layer built to answer it

Rather than argue, an extraction layer was built: twelve modules producing one artifact per question, every row carrying its extraction method and an evidence class drawn from a fixed vocabulary — OBSERVED, DERIVED, VLM_INTERPRETATION, INFERENCE.

The discovery that made this cheap

The recorder had **already captured full screen text** on 939 events, containing complete queue tables, record detail panes, **reference-document bodies including their rule text**, and outcome messages. A pipeline originally scoped to send thousands of screenshots to a vision model was answerable from text already in the data. That pipeline was cancelled: **0 images, 0 model calls, $0.00 spent**, with the capability retained behind a fail-safe interface so that *not spending* remained a measured decision rather than an absence.

Extracted, all from telemetry text and with no model calls: **3,011 records** with their column headers and field values, **112 documents** segmented into **418 clause records** (365 of them long enough to be retrievable units) with parsed rules, **549 outcomes**, **216 queue tables**, and a per-capture state and coverage record.

10.3 The four hypotheses, 44 tests

Eleven modules × four hypotheses. Permutation nulls where a null model is meaningful, 10,000 draws, seed fixed at 7.

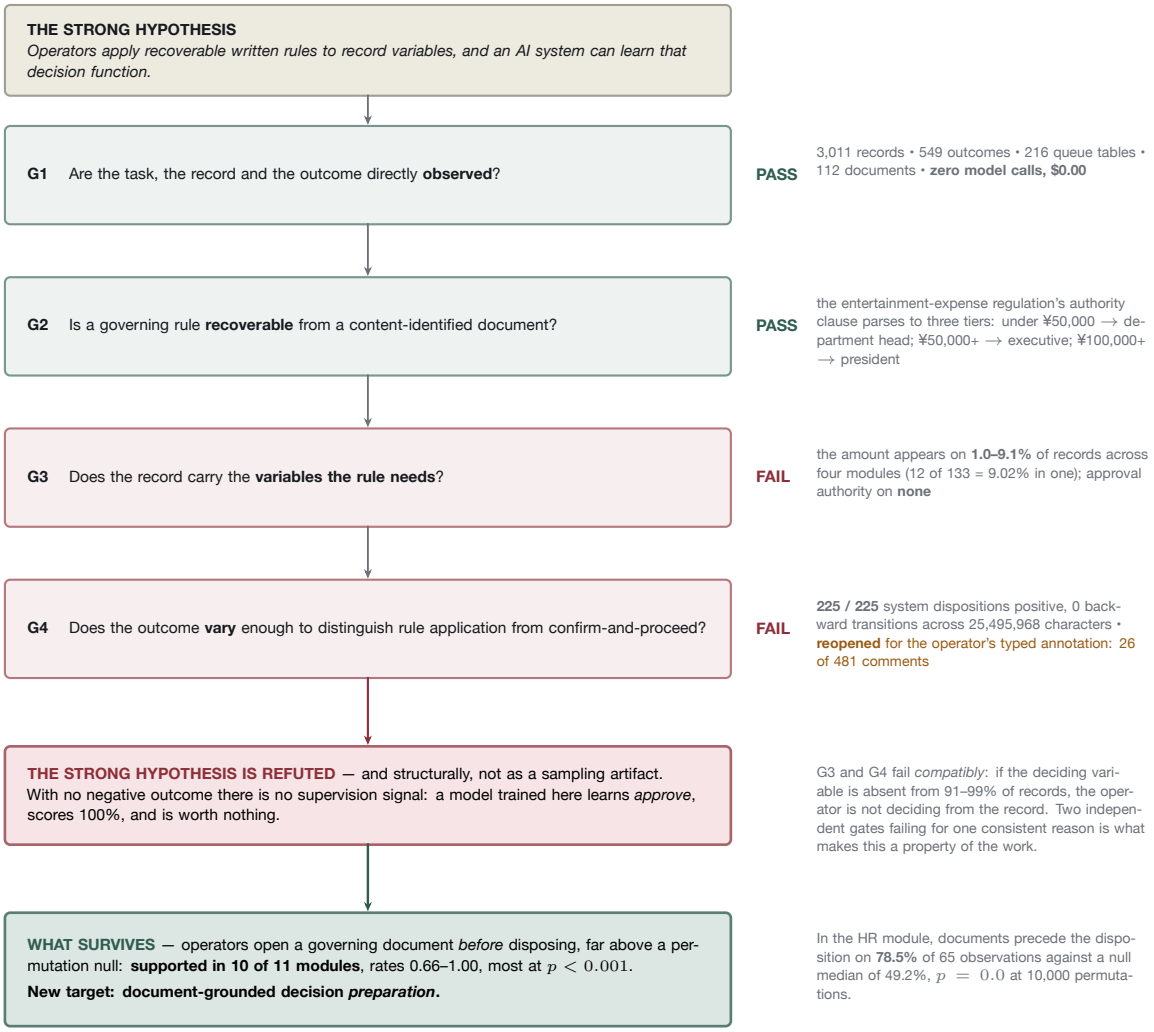


Figure 18. The four-gate test, and what survived it. Two gates pass and two fail — and the two failures corroborate each other, which is what makes the refutation a property of the work rather than of the sample. The survivor is not a consolation prize: document-before-disposition is supported in 10 of 11 modules against a permutation null, and it is the claim the delivered product rests on.

Hypothesis	Test	Result
Rule → disposition	compare rule prediction against observed outcome polarity	**8 untestable • 3 insufficient**
Difficulty → effort	Spearman correlation between record amount and seconds-to-disposition, against a permutation null	**3 refuted • 8 insufficient**
Document → disposition	observed document-before rate against a null that places a document event uniformly within the same execution span	**10 supported • 1 refuted**
Record → rule inputs	set comparison between rule variables and observed record fields	**4 refuted • 7 insufficient**

Table 21. 44 tests over 11 modules. The evidence floor, the permutation count and the seed were all fixed before inspection.

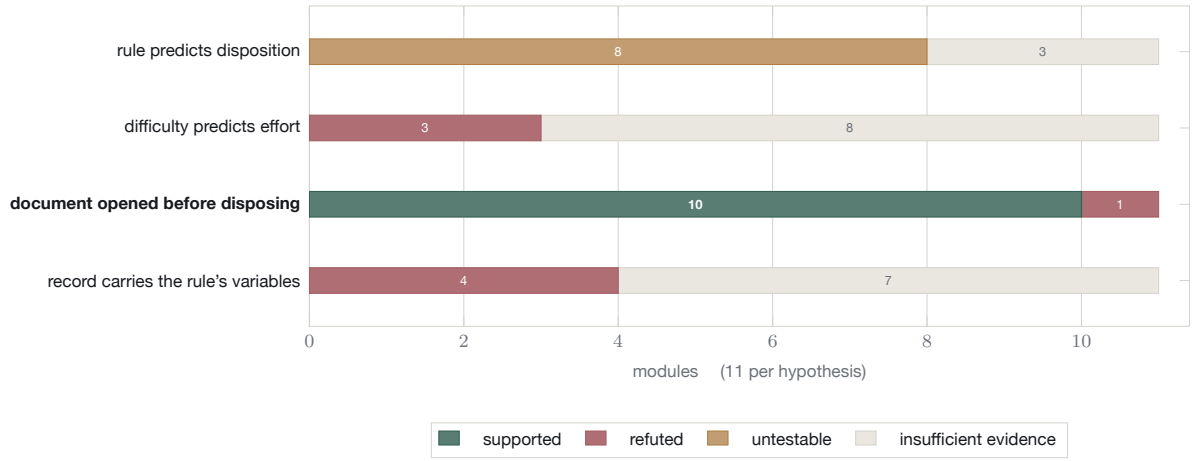


Figure 19. One hypothesis survives, and it is not the one that would have been convenient. Outcomes across 11 modules for each of the four hypotheses. *Untestable* is a stronger statement than *refuted*: in eight modules every observed disposition has the same polarity — 33 of 33 positive in one, 36 of 36 in another, 31 of 31 in the invoice module — and a rule cannot be shown to predict a constant. Recording that as "untestable" rather than "100% accurate" is the difference between a result and a fabrication.

Difficulty-predicts-effort is refuted where it could be tested. Spearman $\rho = -0.2737$ ($p = 0.2317$) in the HR module and $\rho = -0.0668$ ($p = 0.7816$) in the invoice module, against a null whose median $|\rho|$ is 0.1564 and whose 95th percentile is 0.4481. Larger amounts are not slower to process.

Record-carries-the-inputs is refuted. In one module the amount appears on 12 of 133 records (**9.02%**) and approval authority on **zero**. Across the tested modules the deciding variable is present on **1.0% to 9.1%** of records.

Document-before-disposition is the survivor, and it is strong. In the HR module, documents precede the disposition on **78.5%** of 65 observations against a null median of 49.2%, $p = 0.0$ at 10,000 permutations. Supported in 10 of 11 modules at rates 0.66–1.00, most below $p < 0.001$.

10.4 The outcome-variance sweep, and the error I made

Before letting the outcome gate fail permanently, every free signal was swept: all extracted text, all event types, element labels, URLs and screenshot filenames — **25,495,968 characters over 20,477 events** — for any negative, exceptional or reversal state, plus backward or lateral status transitions and any record appearing twice with different terminal states.

The first pass produced a clean null and closed the gate. It was wrong.

It had examined every place a *system* outcome appears and never read what the operator **typed**. Of 2,021 raw keyword hits, 836 were user-interface affordances — the comment box's own placeholder text — 76 were infrastructure errors unrelated to business outcomes, and 0 were candidate business outcomes. That much held. **But the typed-comment field had not been read at all.**

Reading it: **481 distinct operator-typed comments**, of which **26** carry "discrepancy found", "requires attention" or "requires confirmation", against 36 "no discrepancy" and 46 "normal". Every flagged record was *still dispositioned positively* — an operator would type "…requires attention. Purchase order confirmed, registration complete."

The correction stands in the history as a separate commit alongside the wrong one, unamended. Its consequence is the single most useful output of this layer: **the learnable target is the operator's annotation, not the system's decision.** The outcome variance exists; it is in the wrong field, and it describes what the operator *noticed*, not what they *decided*.

10.5 The four-gate verdict

Gate	Criterion	Verdict	Evidence
G1	Task, record and outcome are directly observed	**PASS**	3,011 records • 549 outcomes • 216 queue tables • zero model calls
G2	A governing rule is recoverable from a content-identified document	**PASS**	112 documents; the entertainment-expense regulation's authority clause parses to three tiers: under ¥50,000 → department head; ¥50,000+ → executive; ¥100,000+ → president
G3	The record carries the variables the rule needs	**FAIL**	amount on 1.0–9.1% of records across four modules; approval authority on none
G4	The outcome varies enough to distinguish rule application from confirm-and-proceed	**FAIL** for the system disposition • **REOPENED** for the operator's annotation	225/225 positive, 0 backward transitions • 26 of 481 typed comments carry a discrepancy marker

Table 22. The four-gate verdict. G3 and G4 fail *compatibly*, which is what makes the refutation structural: if the deciding variable is absent from 91–99% of records, the operator is not deciding from the record.

One contradiction, correctly reported as an anecdote. A single record — entertainment expenses at ¥123,224 — sits in a band the regulation routes to president approval, while the queue it sits in is a managerial-approval queue. $n = 1$. It supports no conclusion, and it is labelled as illustrative wherever it appears, including in the product.

11 The pivot: from autonomous decision to AI-assisted preparation

11.1 Why this is a narrowing, not a failure

The experiment did not remove the AI problem. It replaced an ill-posed one with a well-posed one.

Predict the disposition was ill-posed: the dependent variable is constant, so the task has no learnable content and no way to be validated. *Retrieve the governing document, localise the clause that applies to this record, check the record against it, and explain the result* is well-posed — every step is an inference step, each has a definable output, and three of them are measurable against observed behaviour.

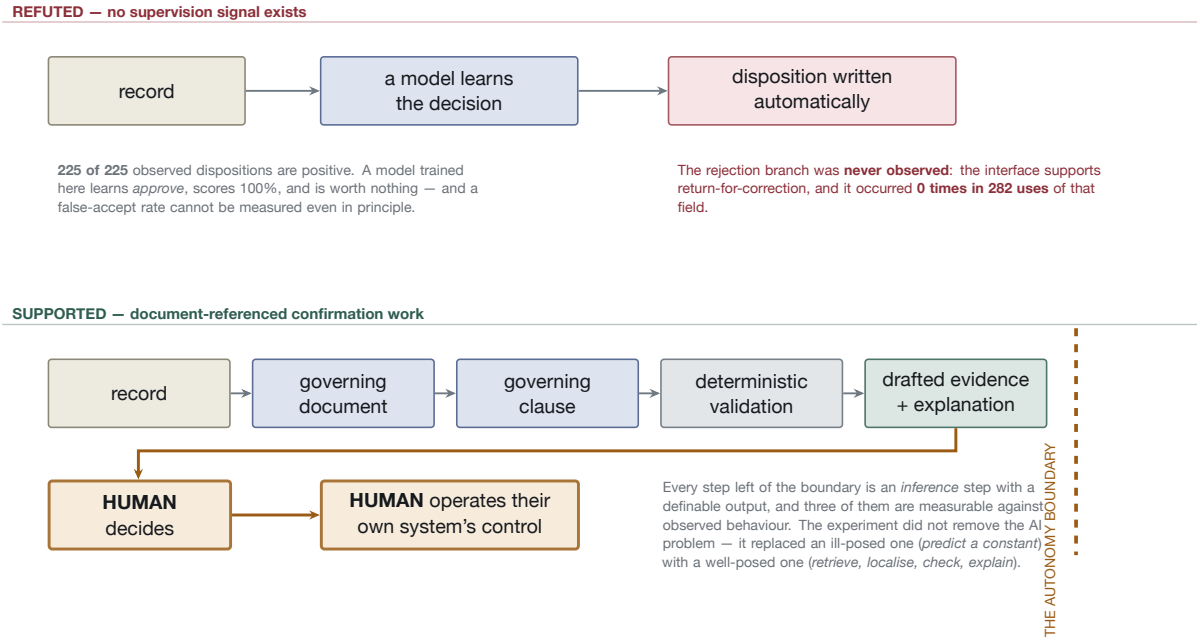


Figure 20. What was abandoned, and what replaced it. The refuted design put a model between the record and an irreversible write. The supported design puts inference everywhere *except* there: retrieval, clause localisation, deterministic checking and drafting all sit left of the autonomy boundary, and the disposition sits right of it. This is the central architectural argument of the engagement, and every element of it is a consequence of a measurement in § 10.

The evidence supports one further claim about this: the operators are doing **document-referenced confirmation work**. They retrieve a record, locate and open the document that governs it, and record a disposition, and they do the middle step far above chance in 10 of 11 modules. The expensive part of that is not the decision — it is finding the document, locating the clause, and assembling what the clause needs.

11.2 The safety argument for the boundary

The boundary is drawn at **reversibility**, and the evidence makes the case concretely.

Argument	Evidence
There is no negative class	225 of 225 dispositions are positive, so a false-accept rate cannot be measured even in principle on this data. A system that cannot be evaluated on its errors must not be allowed to make them.
The rejection branch was never observed	the comment box's own placeholder reads "enter an approval comment or a reason for return" — so return-for-correction is a supported action in the real interface, and it occurred **zero times in 282 uses** of that field across the dataset. A system built here has never seen the branch the interface is built around.
Failures would be invisible	a wrong preparation is visible to the operator and costs a second. A wrong disposition is a statutory write that nobody notices.
The external benchmark agrees	42.7% task success for the strongest agent on enterprise form filling (§ 7.4).
The domain insists on it	Japanese practice holds that verification survives automation.

Table 23. The five arguments for placing the boundary at reversibility. Where the evidence *does* license autonomy, the system takes it: it prepares 120 records unattended, without asking. The boundary is drawn from measurement, not from caution as a style.

12 ProcAssist — the delivered system

ProcAssist

A panel mounted **inside** the business system the operator already uses. For the record on screen it resolves the governing regulation, shows the governing clause verbatim, evaluates the record against it, reports what is missing, and drafts the comment the operator would otherwise type — **with the judgement slot empty**.

The operator chooses a judgement and clicks their own system's submit button. The placement is a product decision taken from the data: the measured cost is *leaving the record* — switching to a word processor to find a regulation and switching back — not the typing. A separate application would have removed one context switch and added another.

WHAT IT DOES

Resolves (module, record type) to a governing regulation • localises and quotes the governing clause with its character span • evaluates declared rules against record fields • reports what the rule needs and the record lacks • drafts the operator's comment • prepares an entire queue unattended • logs every suggestion and every human response to a hash-chained audit trail.

WHAT IT DELIBERATELY DOES NOT DO

Click a disposition — **no code path can** • preselect a judgement • emit a confidence score • guess a document when the lookup misses • fabricate an amount a screen does not show • generate free-text comments with a model • make any absolute time or cost claim.

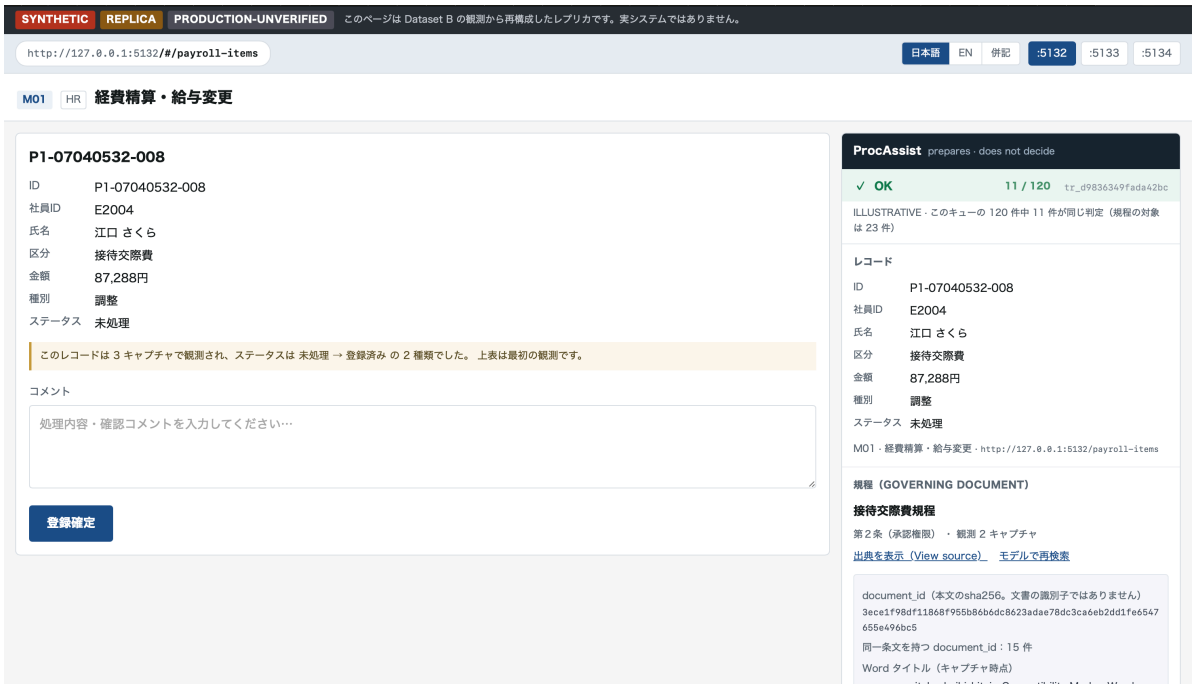


Figure 21. ProcAssist in place. The panel sits inside the operator's own system rather than beside it, because the measured cost is leaving the record — not the typing. Note that the record, the resolved regulation, the quoted clause, the checks and the draft are all visible without scrolling, and that the host system's own submit control is untouched.

12.1 The end-to-end journey

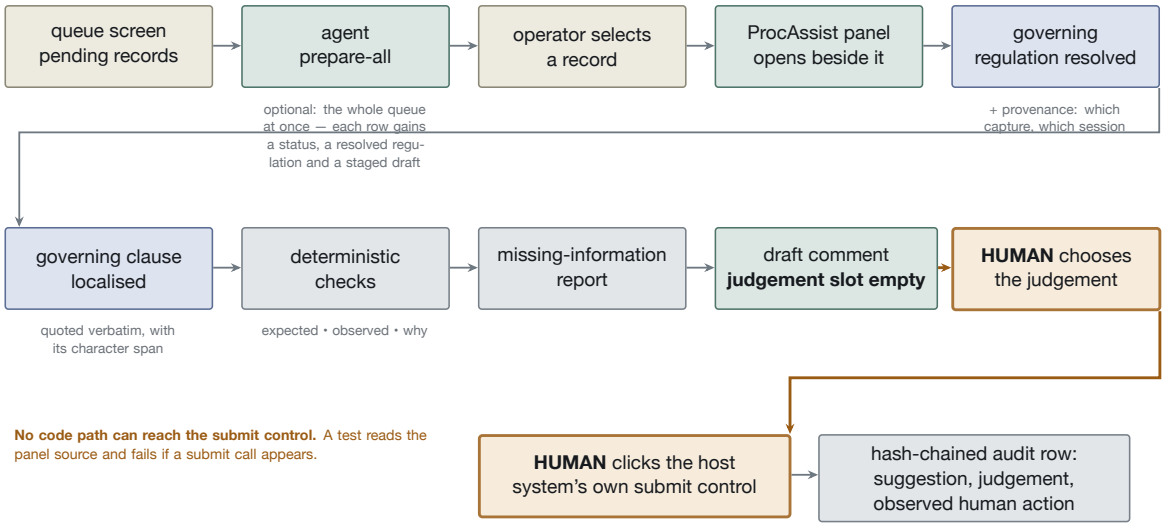


Figure 22. The operator journey, with the two human steps drawn in the autonomy convention. Everything up to the draft is prepared by the system, including an optional unattended pass over the whole queue. The judgement and the submit are the operator's, and the audit row records both what was suggested and what the human actually did — which is the measurement that would later justify moving the boundary.

12.2 Scope: a shared foundation, one module deep

The client's brief explicitly made the scope decision part of the ROI question — one specialised tool, or a shared foundation with per-process definitions. The answer delivered is **a shared foundation with three declared queues**, because one queue cannot demonstrate that a foundation is a foundation and twelve cannot be finished in the budget.

Queue	Records	What it demonstrates
HR expense settlement	120	OK / FLAG — the one queue where the corpus's single parsed authority rule and the amount it needs appear on the same screen
Finance invoice approval	108	MISS — no regulation in the corpus governs an invoice amount, on all 108
Finance managerial expense approval	24	REQUIRES_REVIEW — that screen has no amount column at all

Table 24. Three declared queues, each demonstrating a different status. Three rather than two is an evidence-forced decision, not padding: the corpus contains **exactly one** parsed threshold rule, and no record in the first two queues *lacks* an amount, so the missing-input status was unreachable without a third screen — and fabricating a gap to demonstrate a status was not an option.

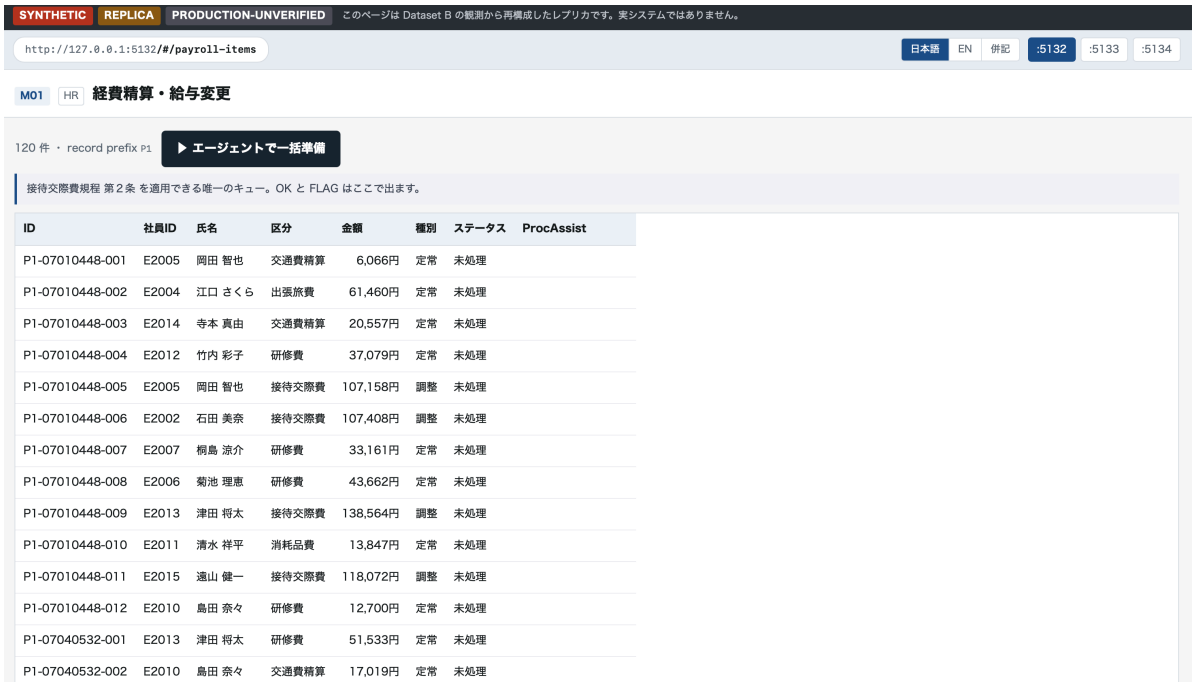


Figure 23. The starting state: the operator's own queue, unassisted. Nothing is resolved; the three honesty labels (SYNTHETIC / REPLICA / PRODUCTION-UNVERIFIED) are visible in the page chrome, as they are at all times. The column headers are reproduced verbatim from the source telemetry, not paraphrased.

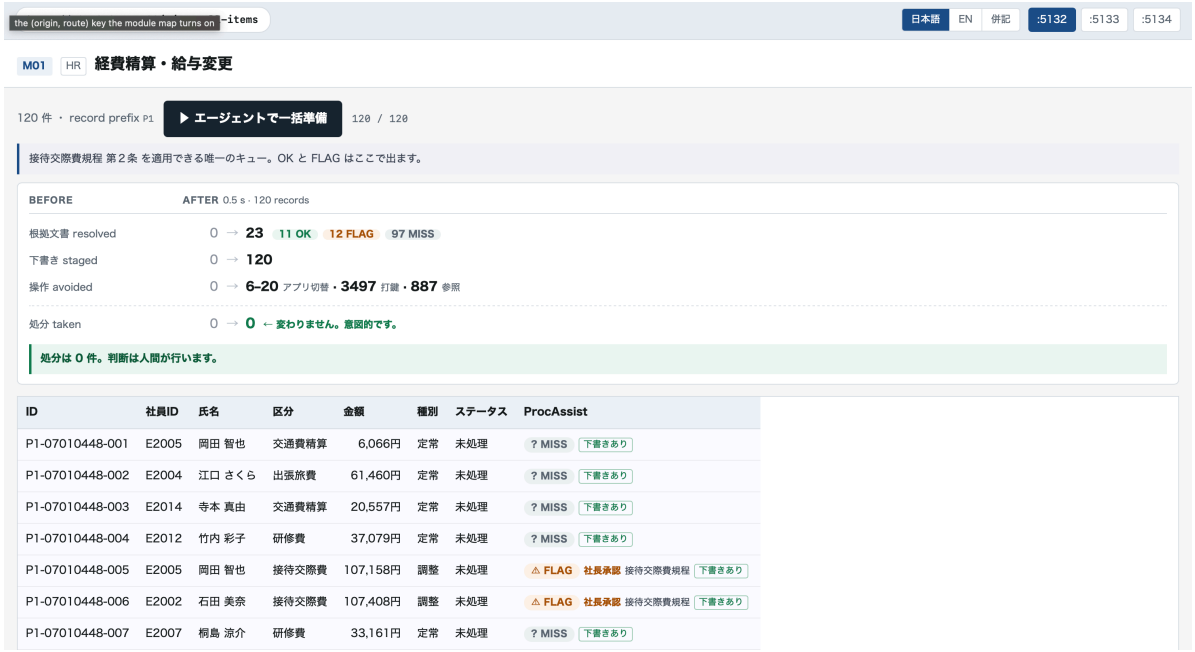


Figure 24. The queue after preparation, with the decision boundary intact. All 120 rows have been prepared, 23 governing documents are resolved and 120 drafts are staged. The summary strip reports 6–20 avoided application switches, 3,497 keystrokes and 887 look-ups, while the final counter remains **0 dispositions taken**.

12.3 What it deliberately does not do

Not done	Why
Click a disposition	no negative class exists; the action is irreversible. **No code path can**: a test reads the panel source and fails if a submit call appears
Preselect a judgement	225/225 positive dispositions already suggest rubber-stamping; a pre-filled judgement would deepen exactly that failure
Emit a confidence score	the document-to-record mapping is deterministic; a number attached to it would be invented precision
Guess a document when the lookup misses	the best deterministic lookup errs on 35.8%; a plausible wrong regulation is worse than an honest refusal
Fabricate an amount a screen does not show	the missing-input status is the correct answer
Generate free-text comments with a model	the observed comment is a fixed template with one variable slot — the judgement — and that slot belongs to the human
Make any absolute time or cost claim	waiting times were compressed in the capture environment

Table 25. Seven capabilities deliberately withheld, each with the evidence that withheld it. This table is the product specification as much as the feature list is.

13 High-level design

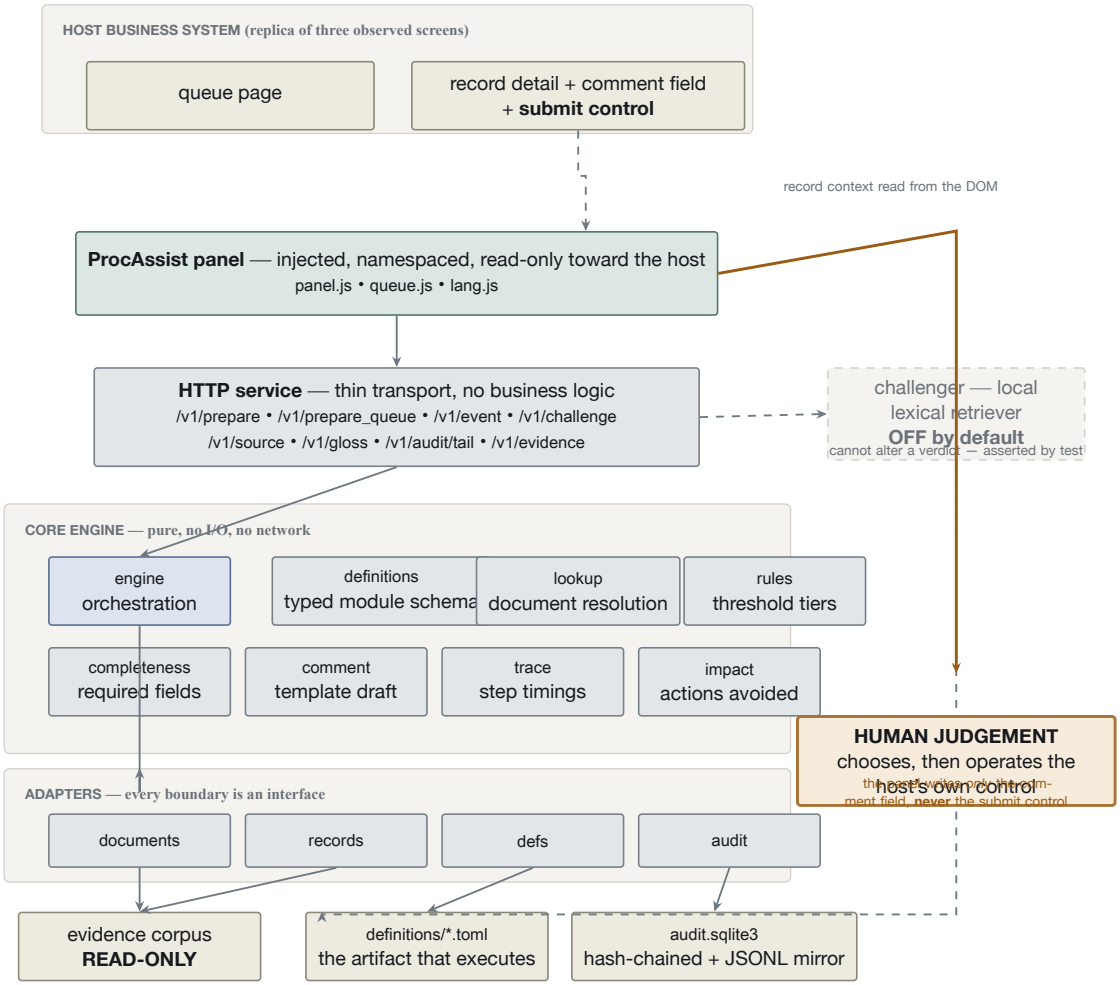


Figure 25. ProcAssist, as implemented. Every component named here exists in the delivered package; nothing in this diagram is conceptual. The core is a pure library with no I/O and no network, which is what makes every integration surface thin. Two elements carry the report's colour convention with intent: the retrieval challenger is drawn as a ghost because it is off by default and cannot alter a verdict, and the human-judgement node is the only place in the architecture where an irreversible action occurs.

13.1 Component responsibilities

Component	Responsibility	Failure behaviour
Core engine	orchestrate preparation for one record; roll individual checks up into one status	raises on any internal contract violation; never degrades silently
Definitions	typed schema for a module: record shape, required fields, governing document reference, threshold rule reference, comment template, authority ladder and ceiling, host origin and route	**fails at load time, not at request time** — a malformed definition prevents startup
Lookup	resolve (module, record type) → governing regulation	**miss is a first-class return value**, never a ranked guess
Rules	evaluate threshold tiers against a record variable; report expected, observed, reason and the tier semantics applied	missing variable → missing-input status, never a default
Completeness	compare required fields against present fields	absent field is reported, not imputed
Comment	render the observed template with record slots filled	asserts the judgement is empty; raises otherwise
Trace	record each engine step, its mechanism and its measured duration	absent trace degrades the panel display only
Impact	convert the measured action budget into per-record avoided actions	missing measurement → the section is omitted, not estimated
Document adapter	index the corpus; return exact clause excerpts with provenance, character span, capture and session identifiers	unknown document identifier → miss
Record adapter	load records; key them on record prefix; parse Japanese amounts	unparseable amount → treated as absent
Audit	append-only log, hash-chained, with a JSONL mirror	write failure surfaces; the chain is independently checkable
HTTP service	thin transport; no business logic above the core	validation errors return structured failures
Panel	render the result; capture the human's response; write **only** the comment field	cannot reach the submit control by construction
Replica	reproduce three observed screens with real element identifiers and honesty labels	labelled SYNTHETIC / REPLICA / PRODUCTION-UNVERIFIED in the page chrome at all times
Challenger	run a local lexical retriever *beside* the lookup for comparison	**cannot alter a verdict**; a test asserts the status, document and clause are unchanged after it runs

Table 26. Component responsibilities and, crucially, failure behaviour. The right-hand column is the design: every component has a defined way of refusing, and none of them degrades quietly.

13.2 The definition file is the artifact that executes

Every rule the engine applies is a line in a declaration file, with its evidence in a comment beside it. This is the governance property that makes the system reviewable: an auditor reads three files, not a codebase, and each value carries the Dataset B observation that produced it.

> **The tiers are not restated in the declaration**
>
> They are read from the document corpus at runtime, so the declaration and the corpus cannot drift apart — the file names *which* clause governs, and the clause supplies the numbers.

13.3 Status vocabulary, and the absence of a score

Four statuses, ordered by severity for roll-up: MISS outranks REQUIRES_REVIEW outranks FLAG outranks OK.

Status	Meaning	Operator action implied
OK	exactly one tier of the governing clause matches; required fields present	proceed with the normal judgement
FLAG	the authority the clause requires exceeds this queue's declared ceiling	look before proceeding
REQUIRES_REVIEW	a rule governs this record but an input it needs is not on the screen	obtain the input
MISS	no governing rule is established for this record type	the tool has nothing to contribute; proceed unaided

Table 27. The four statuses. A single number would invite trading these apart — "0.7 ok" — when three of the four mean *stop and look* and only one means *the rule held*.

13.4 Integration surfaces

The core is a pure library, so every surface is thin. This is the architectural answer to the open question in § 7: whether the client's systems expose a bulk-import or API path is **unverified**, so hard-wiring one execution modality would bet on the client's systems being worse than they probably are.

Surface	Delivered	Consumer
Injected panel	**yes**	the operator, inside their own system
HTTP API, typed request and response	**yes**	the client's own portal, or any client
Server-rendered evidence page	**yes**	a reviewer or auditor
Audit reader	**yes**	governance, and the residual analysis that decides whether a model is warranted
Command-line / programmatic access to the same core	**yes** (the core is importable; the audit reader is documented)	operations, evaluation harness
Workflow-platform node, agent tool surface	**not built** — the HTTP contract is the integration point	deferred deliberately
Direct core-system API or bulk import	**not built — existence unverified**	would change the recommendation if confirmed

Table 28. Integration surfaces delivered and deliberately not delivered. The last row is the highest-value open question in the engagement, and the architecture is shaped to keep its answer available.

14 Low-level design

14.1 Request lifecycle

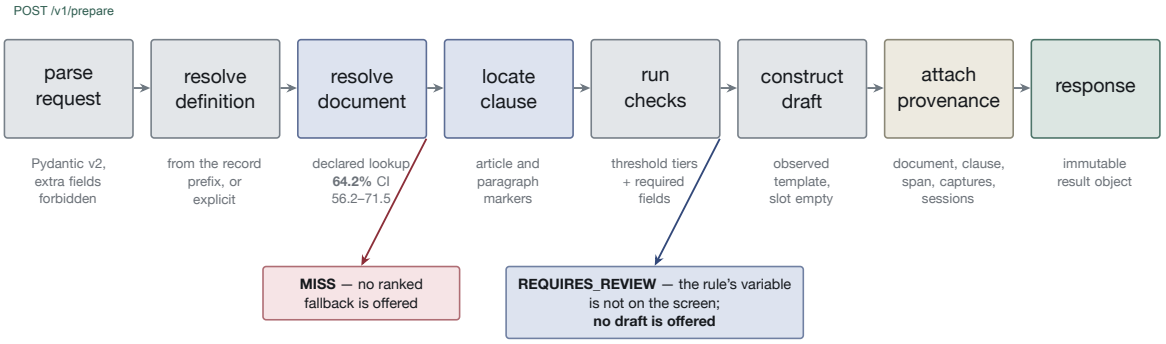


Figure 26. One request, eight steps, two ways to refuse. The lifecycle of POST /v1/prepare. Two of the eight steps are inferential — document resolution and clause localisation — and both are drawn in the AI convention; the rest are deterministic by choice. The two refusal paths are first-class returns, not error handling: neither produces a ranked guess and the missing-input path produces no draft at all.

14.2 Request and result schema

REQUEST

```
POST /v1/prepare
{ "record_id": "P1-07106250-010",
  // optional; inferred from the
  // record prefix when omitted
  "definition_id": "hr_expense_settlement",
  "trace": true }

POST /v1/prepare_queue
{ "definition_id": "hr_expense_settlement",
  "limit": null }
```

THE INVARIANT THAT CARRIES THE PRODUCT

```
def model_post_init(self, _context):
    if (self.draft is not None
        and self.draft.judgement is not None):
        raise ValueError(
          "PrepareResult.draft.judgement "
          "must be None: the engine "
          "never decides")
```

The result object is immutable and forbids extra fields. This method is the reason the judgement slot cannot be filled by accident anywhere in the system.

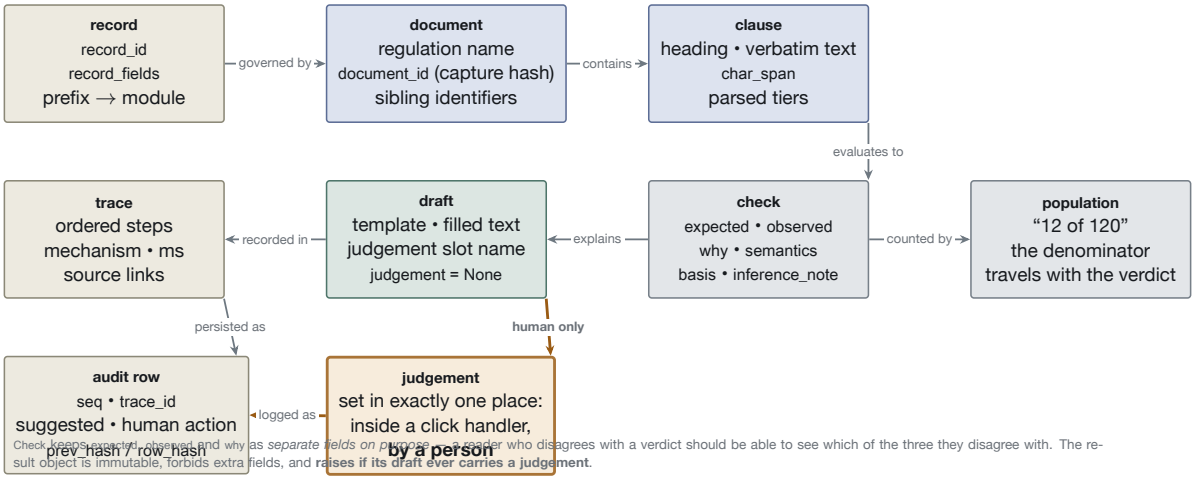


Figure 27. The result contract as a data model. Every relation is enforced by the schema. The population entity is worth noting: a single flagged record invites a reader to treat it as an anecdote, so the denominator travels with the verdict and the panel renders "12 of 120".

14.3 Endpoints

Method	Path	Purpose
POST	/v1/prepare	prepare one record
POST	/v1/prepare_queue	prepare every record on a queue; returns per-row results and before/after totals
POST	/v1/event	record a human response — draft used, suggestion overridden, judgement chosen, host submitted
POST	/v1/challenge	run the local retriever beside the lookup and return both, with the verdict between them
GET	/v1/source/{document_id}/{clause_id}	the verbatim source excerpt and its provenance
GET	/v1/gloss	the committed Japanese→English screen gloss
GET	/v1/audit/tail	the most recent audit rows
GET	/v1/evidence	the measured results the evidence page renders
GET	/, /{route}, /{route}/{record_id}, /_evidence	portal, queue, detail and evidence pages

Table 29. Eight service endpoints and four server-rendered pages. Nothing in the request path reaches a model, and there is no endpoint that performs a disposition.

14.4 Definition format

Declarative, in a standard-library-parsable format — deliberately not YAML, which would have required a fifth dependency. Abridged, with the real evidence comments retained.

```
definition_id = "hr_expense_settlement"
module_id = "M01"
record_prefix = "P1"

# Verbatim from the queue table's own headers on 120 distinct records. Keyed on
# record_prefix, not module id: the URL was measured wrong about the screen in
# 21% of captures, while prefix -> module holds at purity 1.00.
column_headers_ja = ["ID", "社員ID", "氏名", "区分", "金額", "種別", "ステータス"]

# Element identifiers, verbatim from click events: this field used 472 times,
# this button 240.
note_element_id = "pi-note"
submit_element_id = "btn-pi-ok"

[host]
origin = "http://127.0.0.1:5132"
route = "/payroll-items"

[governed_by] # which records this regulation governs
field_ja = "区分"
equals_ja = "接待交際費"

[document]
regulation_name_ja = "接待交際費規程"
clause_id = "c002"
clause_heading_ja = "第 2 条（承認権限）"
document_id = "3ece1f98df11..." # a capture-window content hash,
                                # NOT a document identity
identity_note_en = "15 captures carry this clause under 5 different window titles"

[threshold] # the tiers are NOT restated here; they are read
rule_id = "3ece1f98df11:c002" # from the corpus, so this file and the
variable_ja = "金額" # corpus cannot drift apart
source_field_ja = "金額"

[authority]
ladder_ja = ["部門長承認", "役員承認", "社長承認"] # QUOTED from the clause - observed
ceiling_ja = "役員承認" # DECLARED policy parameter
basis = "INF"

[comment]
template_ja = "経費精算確認済み。費目：{expense_type　金額：}{amount。}{judgement。}"
judgement_slot = "judgement"
```

```
observed_n = 142 # 142 of 223 typed values on this queue take this shape
```

S10 | SCREENSHOT TO BE INSERTED

Capture: A terminal showing curl against POST /v1/prepare for record P1-07106250-010 with "trace": true, and the pretty-printed JSON response

State: the full JSON response, with "judgement": null and the provenance block visible

screenshots/S10.png

Figure 28. The same result object the panel renders, available over HTTP. The integration surface is the typed contract, not the panel: the response carries the status, the document and clause with provenance, the checks with expected/observed/why, the draft with "judgement": null, and the per-step trace with measured timings.

14.5 Document representation and clause localisation

Each captured document carries its body, a clause segmentation, and parsed rules. Clauses are split on Japanese article and paragraph markers (第N条, 第N項), each retaining its heading, verbatim text and character span within the captured body. Threshold rules parse into tiers with an operator, a yen-normalised value and a consequent.


```
{ "rule_kind": "threshold", "clause_id": "c002",
  "raw_text_ja": "回あたり万円未満：部門長承認。万円以上：役員承認。万円以上：社長承認。15510",
  "parsed": { "variable": "金額", "unit": "JPY", "tiers": [
      { "raw": "万円未満5", "operator": "<", "value_jpy": 50000, "consequent": "部門長承認" },
      { "raw": "万円以上5", "operator": ">=", "value_jpy": 50000, "consequent": "役員承認" },
      { "raw": "万円以上10", "operator": ">=", "value_jpy": 100000, "consequent": "社長承認" }]}}
```


A finding about document identity that shaped the design

The document identifier is a content hash of the *captured window*, not of a document: **15 distinct identifiers carry this same clause under 5 different word-processor window titles**. The regulation *name* is the identity that means something; the hash says which capture the excerpt on screen came from. The system therefore keys on the regulation name, carries the hash as provenance, and surfaces the sibling identifiers rather than pretending to a clean document identity it does not have.

14.6 Validation logic, and one correction

Tiers are evaluated with **most-specific-matching-tier** semantics, and this was a correction to a first implementation.

The clause's tiers overlap as written: ¥135,181 satisfies both "¥50,000 or more" and "¥100,000 or more". The first implementation reported that overlap as the verdict — the clause selects no authority. That is true of the string and wrong about the domain: tiered Japanese approval clauses of this form are read as **bands**, with the most specific tier winning. The engine now applies band semantics, the applicable authority for ¥135,181 is president approval without qualification, and **the literal overlap survives as an inference note on the check rather than as the answer**.



Figure 29. Every statement resolves to a source. The panel names the governing regulation, quotes the clause verbatim in Japanese — it is *never* machine-translated at runtime — and links to an excerpt carrying the document identifier, clause identifier, character span, and the captures and sessions it was seen in. The reader can always answer: *which document, which clause, which capture?*

The one policy knob in the system, and it is declared rather than measured

A FLAG means the authority the clause requires exceeds the **declared ceiling** of the queue the record sits in. The ladder is quoted from the clause and is observed. The ceiling is not — and we measured before declaring it: of 8 distinct entertainment-expense settlements observed on that queue, **5 fall in the executive-approval band** (¥60,936–¥98,018) and **3 in the president-approval band** (¥118,189 / ¥123,224 / ¥135,181), and **no observed field distinguishes them** — all 23 such records share the same category, with no exception marker and no badge. Where a settlement clerk's authority stops is therefore the client's question, not the corpus's. It is one line of the declaration, the panel renders its inference tag on screen, and the 11-ok / 12-flag split out of 120 follows from it. **A reviewer must not read that split as a measurement**, and the product says so.

S05 | SCREENSHOT TO BE INSERTED

Capture: The panel's check list for a FLAG record, with the expected / observed / why fields and the inference tag visible

State: the FLAG check with all three fields, the semantics and basis tags, and the "12 of 120" population label

screenshots/S05.png

Figure 30. Deterministic validation, shown in three separable parts. A reader who disagrees with a verdict can see which of expected, observed and why they disagree with. The inference tag beside the check marks the declared ceiling as a policy parameter rather than a measurement, and the literal tier overlap is carried as a note beside the verdict rather than folded into it.

14.7 Draft generation

The template is reconstructed from the operator's own typed values: **142 of 223** typed values on that queue take exactly the delivered shape, with the remaining 81 belonging to the module's other half. Slot resolution is declarative (field:区分, amount:金額). The judgement slot renders empty. Exactly one judgement option is offered, because exactly one was ever observed — a second was not invented to make the interface symmetrical — and it still has to be clicked.

Formatting fidelity is deliberate down to the character

The ideographic space (U+3000) before the amount label, and the yen character as a *suffix*, because there is no yen sign in this module's observed comments. When the human clicks the option, the completed comment is **byte-identical** to a string an operator actually typed, and there is a test asserting exactly that.

14.8 Trace, persistence and configuration

The **trace** records each engine step with its name in both languages, the mechanism it used (lookup, threshold, template), a measured duration in milliseconds, and where relevant a link to the source excerpt.

Persistence is an append-only SQLite table with a JSONL mirror. Each row carries a sequence number, timestamps, a trace identifier, the event type, the definition and module, the record, what was

S06 | SCREENSHOT TO BE INSERTED

Capture: The draft comment in the panel before the judgement is chosen, with the host system's comment field filled and the empty judgement slot visible

State: the filled comment field, the empty judgement slot between the final two characters, and the single available judgement option with its observed count

screenshots/S06.png

Figure 31. The draft, with the judgement slot empty. The staged text reads 経費精算確認済み。費目：接待交際費　金額：135,181円。。 — note the two consecutive full stops, which is where the judgement is missing. The tool fills the host's comment field and nothing else.

suggested, what the human did, and prev_hash / row_hash forming a verifiable chain. The audit trail from a real browser session is in the repository.

Configuration is minimal by design: no secrets, no API keys, no model endpoints. A single environment variable enables the retrieval challenger, which is off by default. There is nothing to configure because there is nothing external to reach.

14.9 Panel interaction and its invariants

The panel is a single script namespaced away from the host's own styles. It reads record context from data attributes on the host element, calls the service, renders, and captures the human's response. Three invariants, each enforced in code and each with a test.

> **1 | It never writes to the host's disposition control.** The submit element's identifier is read into state and never used; there is no click or submit call anywhere in the panel source. A test greps the source and fails if one appears.
>
> **2 | No judgement is preselected.** The chosen judgement initialises to null and is assigned in exactly one place — inside a click handler, annotated in the source as *set here, by a person, and nowhere else*.
>
> **3 | Every staged draft has an empty judgement slot**, including across a whole-queue batch — a test asserts this holds for all 120 records at once.

S07 | SCREENSHOT TO BE INSERTED

Capture: The judgement step: the panel with the single judgement option unselected, and the host system's submit control clearly outside the panel

State: unselected judgement option, the panel boundary, and the host's own submit button in its original position

screenshots/S07.png

Figure 32. Where the system stops. The judgement option is rendered but not chosen, and the host's own submit control sits outside the panel's reach. Nothing the tool did can produce the success message; only the operator's click can.

14.10 Error handling

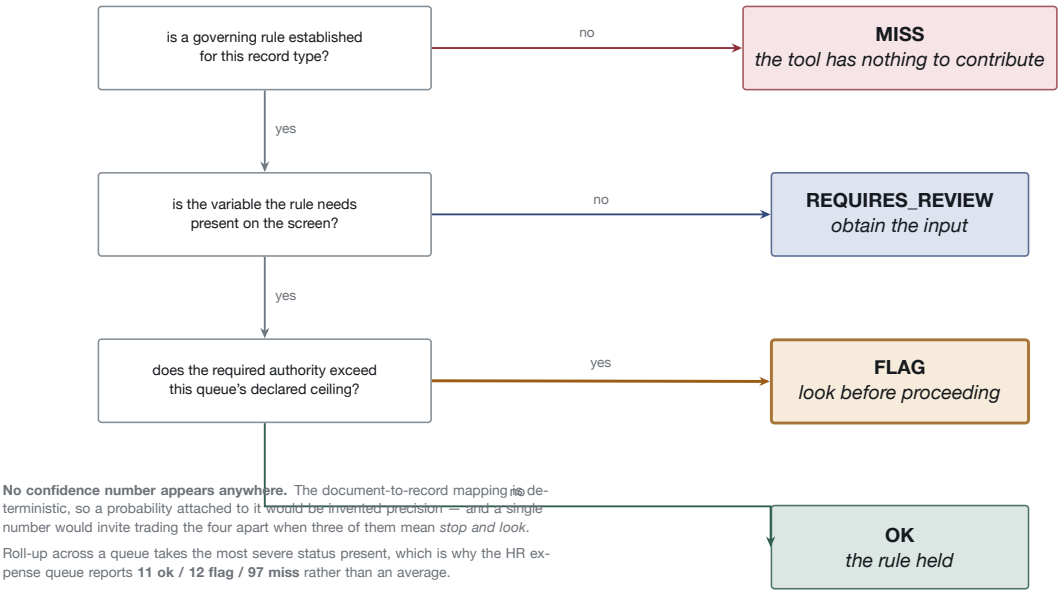


Figure 33. When each status is produced. Three deterministic questions in a fixed order, with severity increasing away from ok. No confidence number appears anywhere in the product, because the mapping being evaluated is deterministic and a probability attached to it would be invented precision.

Condition	Behaviour
Malformed definition	**startup fails** — never a request-time surprise
Unknown record	structured error; no partial result
No governing document	`MISS`, with the reason in both languages; no ranked fallback
Rule variable absent from the screen	`REQUIRES_REVIEW`; no default, no imputation, and **no draft offered**
Unparseable amount	treated as absent, which routes to `REQUIRES_REVIEW`
Unknown document identifier	`MISS` rather than a nearest match
Record seen in several states	the first observation is shown **and the rest are disclosed on the page**
Retrieval challenger disagrees	both answers shown; the lookup remains authoritative; a test asserts the verdict is unchanged
Audit write failure	surfaces; the chain is independently verifiable afterwards

Table 30. Error handling. Every row is a way of refusing rather than a way of degrading — which is the same principle the inference layer learned the hard way in § 5.

15 Where the intelligence is

The most common misreading of a system like this is that it is "not really AI". The accurate statement is that **inference is placed where evidence supports it and withheld where evidence supports only a guess**. Measurement, not fashion, sets the boundary at every layer.

S08 | SCREENSHOT TO BE INSERTED

Capture: The **REQUIRES_REVIEW** state on the managerial-approval queue, where the screen has no amount column at all — panel showing the missing-information report and *no draft*

State: the **REQUIRES_REVIEW** badge, the missing-input reason in both languages, and the absent draft section

screenshots/S08.png

Figure 34. Missing information is reported, never imputed. The rule governing this record needs a variable the screen does not carry. The system says so, names what is needed, and withholds the draft entirely — because offering a draft here would invite the operator to accept a comment the evidence does not support.

S09 | SCREENSHOT TO BE INSERTED

Capture: The **MISS** state on the Finance invoice queue — a prepared record with no regulation named and no ranked fallback

State: the **MISS** badge, the explicit "no governing rule established" reason, and the absence of any suggested document

screenshots/S09.png

Figure 35. The refusal is the feature. No regulation in the corpus governs an invoice amount, and this is true of all 108 records on the queue. The best deterministic lookup errs on 35.8% of cases; a plausible wrong regulation is worse than an honest refusal, so the panel offers nothing rather than a nearest match.

Table 31. Mechanisms by layer. Learned components are used only where the available supervision and failure cost support them.

Layer	Mechanism	Category	Why this mechanism
Normalisation and surface canonicalisation	deterministic transforms	classical	recorder artifacts must be removed reproducibly
Boundary candidates	deterministic proposals plus a measured reachability ceiling	instrumented classical	separates "cannot be proposed" from "chosen wrongly", making Layer 1 diagnosable
Boundary decision	fitted scorer plus constrained decoder	supervised ML	supervision exists on Dataset A; structural invariants belong in the decoder
Process identity	execution representation $\rightarrow$ similarity $\rightarrow$ kNN graph $\rightarrow$ communities, with k chosen by resampling stability	**unsupervised ML**	production has no labels and an unknown class count; ARI 0.681 / NMI 0.738 at 17 labels
Screen semantics	character 2–3-gram TF–IDF	classical NLP	Japanese is not whitespace-delimited; n-grams avoid an unneeded morphological analyser
Ambiguity handling	a gate that routes rather than guesses	classical	a perfect language-model oracle was worth only $+0.026$ F1 at the full budget
Evidence extraction	parsing of pre-captured screen text by evidence class	classical	the cheaper signal was measured sufficient: 0 model calls, $0.00
Document resolution	declared lookup at 64.2% (95% CI 56.2%–71.5%), with a registered promotion bar and local challenger	**retrieval, evaluated**	the challenger can be observed losing without being allowed to change a verdict
Clause localisation	418 clause records segmented on article and paragraph markers, with character-span provenance	**retrieval / IR**	per-record conditioning is the first place a learned component could earn its place
Rule evaluation	parsed tiers, band semantics, explicit statuses	deterministic	a confidence score would be invented precision
Evidence synthesis	bilingual template with an empty judgement slot	deterministic by measurement	142 of 223 typed values take exactly the delivered shape
Decision	**human**	—	0 of 225 observed dispositions were negative and the action is irreversible

Layer	Mechanism	Category	Why this mechanism
Observability	step trace, hash-chained audit and action-budget model	classical	these measurements determine whether the boundary can later move

15.1 Alternatives considered across the whole system

Table 32. Alternatives rejected, retained or deliberately deferred.

Alternative	Benefit	Measured limitation	Decision
Fully deterministic	auditable and simple	cannot recover unlabelled process identity; structural signatures reach ARI 0.431 at 51 labels versus 0.681 at 17	rejected where learning demonstrably earns its place
Language model for process identity	semantic understanding	perfect-oracle merge reaches ARI 0.478, below the 0.583 clustering already achieved	rejected on a measured ceiling
Language model for boundary adjudication	handles an ambiguous tail	perfect oracle is worth $+0.026$ F1 at full budget and $+0.004$ at the budget used	rejected on a measured ceiling
Language model for missing screen semantics	fills a perceived gap	a linear probe over existing features recovers process code at 0.891 accuracy	rejected: the information is already present
Metric learning from bounded model-generated pair labels	targets the actual objective gap	untested and outside the delivered scope	recorded as the next identity experiment
Dense retrieval and neural reranking	strong on large corpora	365 clauses are about 10 MB of vectors; $n = 148$ cannot resolve a margin below the lookup's 71.5% upper bound	reject at this scale; keep the registered promotion bar
Local lexical retrieval	local, keyless and comparable on short Japanese text	labelled pairs are not recoverable from committed artifacts	delivered as an off-by-default, visibly UNEVALUATED challenger
Vector database	approximate search at scale	three to four orders of magnitude below its crossover; exact search is sub-millisecond	rejected with the scale arithmetic recorded
Language-model comment generation	flexible phrasing	the observed comment is a fixed template beside a statutory write	rejected
Autonomous interface agent	removes the human	strongest enterprise form-filling benchmark reports 42.7% task success; no negative class exists	rejected

Alternative	Benefit	Measured limitation	Decision
Pure RPA	deterministic execution	automates the easy part while interface drift dominates maintenance	rejected as the whole answer; retain as a future execution surface
Human-only workflow	zero system risk	document search and templated typing remain pure overhead	baseline being improved
Vision over screenshots	recovers pixel-only evidence	captured screen text already answered the questions; recorder OCR damaged clause headings	cancelled after measurement: 0 images, 0 calls, $0.00

One consistent rule across the engagement

A language model was sized against a measured alternative five times and adopted once, at design time, where the artifact it produced — not the model — is what executes. The instrumentation needed to revisit each boundary is already present.

16 Retrieval and document grounding

16.1 The corpus and its identity problem

The corpus contains **112 captured documents yielding 418 clause records**: Japanese regulations, procedures, checklists and reference tables. Clauses are segmented on article and paragraph markers and retain their heading, verbatim text and character span.

Document identity had to be settled before evaluation. Keying on a content hash yields **50 labels**, because one regulation captured at two scroll positions produces two identifiers. Keying on the document's first heading yields **17 labels**, close to the 13 reference documents the dataset is stated to contain. Both spaces are reported: hashes form the pessimistic bound and depress every score by roughly 20 points.

16.2 How much supervision exists

Population	Count
Dispositions total	629
With a document opened before the disposition in the same execution	**148 (23.5%)**
Without an observed document open	481
Distinct heading labels	17
Majority-label share	18.9%

Table 33. The supervision available for governing-document retrieval.

Per-module pair counts are: onboarding 33; HR expense 24; contract management 19; invoice approval 17; purchase orders 16; benefits 11; managerial expense 9; payment processing 8; inventory 8; attendance 3. Nine of eleven modules clear the pre-declared floor of $n \geq 8$.

At $n = 148$, a single accuracy estimate has an interval of roughly ± 7.5 points. Retrieval is therefore evaluated **pooled across modules**, never per module. Supervision covers less than a quarter of dispositions, so the product is silent where no governing document was observed.

16.3 Baselines, and the registered bar

All results use leave-one-out evaluation over the 148 pairs with a fixed seed. Leave-one-out avoids spending scarce evidence on a small held-out split.

Baseline	Heading space (17)	Hash space (50)
Random	5.4%	1.4%
Most frequent overall	18.9%	18.9%
Most frequent per module	52.7%	27.7%
(module, record type) → document	**64.2%; CI 56.2–71.5%**	42.6%; CI 34.9–50.6%

Table 34. Governing-document retrieval baselines. The label-space choice is shown rather than hidden because it moves the result materially.

> **The registered promotion bar**
>
> A declared lookup reaches 64.2%. A learned retriever must exceed 71.5% pooled top-1 accuracy before this evidence can distinguish it from that table. The honest delivered answer is the lookup plus an interface and instrumentation that make its residual measurable.

A learned component earns reconsideration if the corpus grows, if the correct document depends on a per-record variable a table cannot express, or if the 35.8% residual is genuinely ambiguous rather than dominated by a better key.

16.4 Why no vector database

Embedding 365 retrievable clauses is approximately **10 MB of vectors**. An exact matrix product and sort are sub-millisecond and have no operational surface. Approximate search begins to solve a real problem around 10^5–10^6 vectors, three to four orders of magnitude away. The crossover and scale-out path are recorded; a vector database is not introduced early.

16.5 The challenger, and the failure mode it exposed

The product includes a local lexical challenger: BM25 over character 3-grams plus n-gram cosine across all 365 clauses of at least 12 characters. Character n-grams suit Japanese without adding a morphological analyser. It runs beside the lookup, never in the decision path, and the package is tested for the absence of network clients, API keys and model-loading imports.

On the flagship record it disagrees. The lookup returns the authority clause; the challenger returns a records-keeping article and captured preamble because surface vocabulary matches the noisiest part of the capture. Three safeguards follow:

1. The disagreement is pinned by a test so a future change cannot hide it.
2. Running the challenger cannot alter status, document or clause; scores never enter the result object.
3. The challenger is labelled UNEVALUATED in both languages. The 148 labelled pairs behind the lookup result are not recoverable from committed artifacts, so no challenger accuracy is claimed.

16.6 Provenance

Every panel statement resolves to a source. Results carry document and clause identifiers, heading, character span, rule identifier, captures and sessions, recorded window titles and sibling identifiers. A dedicated endpoint returns the verbatim excerpt. The reviewer can always answer: *which document, which clause, which capture?*

17 Human-in-the-loop design and safety

17.1 Where the boundary is

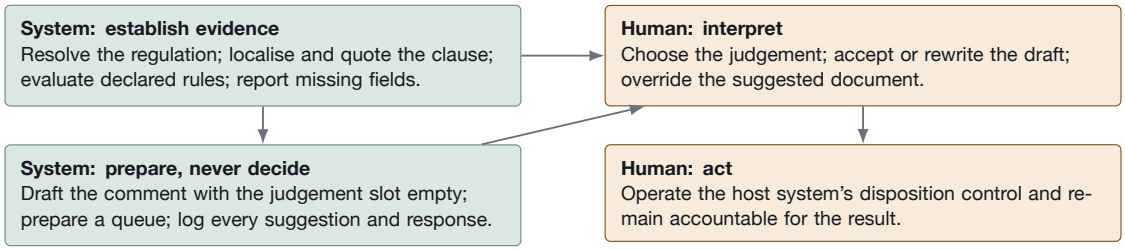


Figure 36. The product boundary. Evidence may be resolved and prepared automatically; judgement and the irreversible action remain human.

17.2 Risks, and what the architecture does about each

Table 35. Safety risks, evidence, mitigations and residual gaps.

Risk	Evidence	Mitigation	Residual gap
Automatic disposition	0 of 225 dispositions negative; action irreversible	no code path sets or clicks submit; a test rejects any submit call	future code could add one, hence the source-reading test
Automation bias	225/225 positive already resembles rubber-stamping	empty, non-defaultable judgement; only observed options; override rate logged	only a pilot can show whether the tool accelerates rubber-stamping
No abstention	lookup error is 35.8%	MISS has no ranked fallback or nearest match	repeated misses may reduce operator trust
Missing information imputed	deciding variables appear on only 1.0–9.1% of records	REQUIRES_REVIEW, no default, no imputation and no draft	operator must source the value
Wrong document surfaced	lookup error and a visibly wrong challenger result	explicit regulation name, source link, alternatives and logged override	a confidently wrong regulation remains possible
Document version drift	15 identifiers carry one clause under 5 titles	dual key on regulation name and hash; unknown hash refuses; siblings surfaced	no version ordering exists
Ambiguous rule	clause tiers overlap literally	declared band semantics; overlap retained as an inference note	domain reading, not corpus fact
Policy input mistaken for measurement	two authority bands occur on one queue without a distinguishing field	ceiling is one declared line, tagged as an inference; report disclaims the OK/FLAG split as measurement	a hurried reader may still misread it
Selector drift	identifiers came from one capture window	identifiers pinned; drift must fail loudly; adaptive recovery rejected	production stability is unverified

Risk	Evidence	Mitigation	Residual gap
Wrong module key	URL is wrong about the screen in 21% of captures	record prefix and screen header are used, never URL alone	prefix evidence covers 4.59% of events
Unobserved rejection branch	return-for-correction used 0 times in 282 field uses	control shown in the replica and labelled never observed	the system has never seen the branch it represents

17.3 The honesty labels

The replica renders SYNTHETIC, REPLICA and PRODUCTION-UNVERIFIED in its page chrome at all times. The honest claim is that the mechanism uses routes, element identifiers and fields found in the client's telemetry. It does *not* claim to have run against production. Stating the second clearly is what makes the first believable.

18 Running the system

Every command below is taken from the repository and is the command the system actually uses.

18.1 Prerequisites

- Python 3.10 or later (the delivered environment uses 3.12).
- The repository virtual environment at .venv, created by make setup.
- **No API key, model download, network access or database server.**
- Playwright is optional. Browser tests and the demo recorder skip cleanly when it is absent.

18.2 Quick start

```
make setup
.venv/bin/python -m procassist.api
```

Open http://127.0.0.1:5132/payroll-items and press ▸ エージェントで一括準備 (prepare all with the agent). The service binds ports 5132, 5133 and 5134 because origin plus route is the module key in the source data.

18.3 The full end-to-end run

Table 36. Ten-step verified walkthrough.

Step	Action	Expected result
1	Open the HR expense queue	120 raw records, verbatim Japanese headers, no resolved evidence, and all three honesty labels
2	Press ▸ エージェントで一括準備	rows resolve in about half a second; the summary reads 0 → 23 documents, 0 → 120 drafts, 6–20 app switches, 3,497 keystrokes, 887 look-ups, and deliberately 0 → 0 dispositions
3	Open FLAG record P1-07106250-010	entertainment-expense regulation and authority clause; ¥135,181 requires president approval; "12 of 120" beside the verdict

Step	Action	Expected result
4	Expand *View source*	verbatim excerpt, document and clause IDs, character span, captures and sessions
5	Expand the agent trace	ordered steps, mechanism and measured timing
6	Press 下書きを使う	host comment field fills with the measured template and an empty judgement slot
7	Click the judgement option	comment completes byte-identically to an observed operator string
8	Click 登録確定 yourself	only the human action produces success
9	Switch 日本語・EN・併記	labels and explanations switch language; authoritative regulation text remains Japanese
10	Open エビデンス / Evidence	four-gate result, retrieval baselines and explicit capability limits

18.4 The other three cases

Record	Expected result
5132/payroll-items/P1-07089771-005	OK — ¥60,936, one tier, executive approval
5133/payroll-items/P6-07010448-004	MISS — no governing regulation and no ranked fallback; true for all 108 invoice records
5133/leave-applications/P9-07084298-006	REQUIRES_REVIEW — screen has no amount column and no draft is offered

Table 37. Cases covering the three non-FLAG statuses.

18.5 Optional: watch the retriever lose

```
PROCASSIST_CHALLENGER=1 .venv/bin/python -m procassist.api
```

This adds モデルで再検索 / try the retriever. On the FLAG record it disagrees for the reason analysed in § 16.

18.6 Read the audit trail

```
.venv/bin/python -c "from procassist.adapters.audit import AuditLog; a=AuditLog(); [print(r['seq'], r['event_type'], r['record_id'], '|', r['suggested'], '->', r['human_action']) for
    r in a.tail(3)]; print('chain_ok', a.verify_chain())"
```

18.7 Tests

```
.venv/bin/python -m pytest -q
.venv/bin/python -m pytest tests/test_procassist_*.py -q
.venv/bin/pip install playwright
.venv/bin/playwright install chromium
```

The final two commands are needed only for browser end-to-end tests and the demo recorder.

18.8 Reproducing the earlier layers

```
make run-b
make validate-segments
make build-b
make docs-pdf
```

`make run-b` is the only path that writes the Layer 1 deliverable. Each run records the identity method, output hash and resolved dependency versions.

18.9 Troubleshooting

Symptom	Cause	Fix
Ports already in use	5132–5134 are occupied	free them; the three-port layout is the observed module key
Browser tests skip	Playwright absent	expected; install it only when the browser suite is required
No challenger control	challenger off by default	start with `PROCASSIST_CHALLENGER=1`
Import error	wrong interpreter	invoke `.venv/bin/python`; the pipeline refuses to run without its numerical dependencies
Empty queue	committed evidence corpus absent	restore `build/dataset_b/evidence`; the product treats it as read-only
Impact figures absent	action-budget measurement absent	section is omitted rather than estimated

19 Evaluation

Four evaluations use different populations and metrics. They are reported separately and must not be combined.

19.1 Layer 1 — execution inference (annotated dataset)

Population: 63 annotated sessions and 162,768 events. Protocol: leave-one-session-out.

Table 39. Layer 1 evaluation.

Measure	Result	Reference point
Boundary F1 at 2 s	**0.690** (P 0.680 / R 0.702)	strongest baseline 0.258; **2.67**×
Segment-count ratio	**0.98**	strongest baseline 2.83
Recall against ceiling	0.702 of 0.900; **78% reachable recall**	never quoted as 0.702/1.000
Candidate reachability	**88.0%** of a 90.0% ceiling	block edges alone: 59.9%
Segmental F1 at IoU 0.5	0.768	stricter whole-segment measure
WindowDiff / Pk	0.470 / 0.257	lower is better
Per-session F1	0.683 ± 0.115	includes one truncated session at 0.022
Identity ARI / NMI	**0.681 / 0.738**, 17 labels	structural labeller: 0.431, 51 labels
Homogeneity / completeness	0.751 / 0.725	both improve at every step
Event ARI	**0.297**	strongest baseline 0.250; 1.19×
Interruption stitching	0.337, from 0.122	degenerate arm: 0.857 at ARI -0.001

19.2 Layer 2 — analysis validation (production dataset)

There is no ground truth for Dataset B. Here evaluation means independent corroboration.

Validation	Result
Prior-claim re-audit	42 claims: **16 held, 11 contradicted**; remainder unresolved
Independent review	commissioned against primary sources, not the analysis
Module-key triangulation	three keys over 3,011 records; screen header and prefix agree 2,904/2,904; URL dissents alone in all 552 disagreements
Prefix-map derivation	re-derived from 211 queue captures; all 13 prefixes at purity 1.00
Cross-module-queue hypothesis	directly refuted: 33 mixed-prefix captures, 0 crossing a module boundary
Completeness	a twelfth module was found missing: 11 captures and 132 rows at purity 1.00
Screenshot and screen-text review	corrected route-to-process interpretation

19.3 Layer 3 — hypothesis testing (production dataset)

Forty-four tests cover 11 modules and four hypotheses. Permutation nulls use 10,000 draws, seed 7 and a pre-declared evidence floor of $n \geq 8$.

Hypothesis	Supported	Refuted	Untestable	Insufficient
Rule predicts disposition	0	0	**8**	3
Difficulty predicts effort	0	**3**	0	8
Document opened before disposition	**10**	1	0	0
Record carries rule variables	0	**4**	0	7

Table 41. Layer 3 verdicts. The outcome sweep covered 25,495,968 characters, 20,477 events, 4,746 screenshot filenames and every status transition.

19.4 ProcAssist evaluation

Table 42. Product evaluation. Dashes are retained where evidence does not exist.

Measure	Result	Population	Status
Governing-document top-1	**64.2%**, CI 56.2–71.5%	148 pairs, 17 labels, pooled leave-one-out	measured
Baselines	random 5.4%; overall majority 18.9%; per-module majority 52.7%	same	measured
Pessimistic label space	42.6%, CI 34.9–50.6%	50 hash labels	measured
Top-3 / MRR	—	—	not measured
Clause localisation	—	no labelled clause truth	tested, not scored

Measure	Result	Population	Status
Rule evaluation	exact ¥50,000/¥100,000 boundaries, bands and missing-variable routing	unit tests	tested
Draft fidelity	byte-identical to an observed string	unit test	tested
Judgement slot	empty for every result and all 120 batch records	unit test	tested
Cannot disposition	no submit call in panel source	source-reading test	tested
Challenger isolation	status, document and clause unchanged	unit test	tested
Challenger accuracy	—	pairs unavailable	UNEVALUATED
HR expense queue	120 records; 23 documents; 120 drafts; 11 OK / 12 FLAG / 97 MISS; 0 dispositions	replica	measured
Invoice queue	108 records; 108 MISS; 0 documents	replica	measured
Audit integrity	verifiable chain; browser-session rows committed	replica	verified
Override, acceptance and agreement rates	—	no pilot	not measured
Time saved	—	compressed waiting times	not measurable here
Test suite	**449 passing, 2 skipped**; 366 functions, 79 product tests	repository	tested

20 Expected business impact

20.1 What can and cannot be claimed

Waiting times were compressed in the capture environment. Candidates must be compared with one another rather than assigned absolute duration or monetary value. **No absolute time or cost claim is made in this report or in the product.** The measurable unit is operator actions.

20.2 Observed — the action budget

The budget was measured once over all 20,477 events, keyed by origin and route, with Wilson 95% intervals.

Screen	Switches	Dispositions	Per disposition	95% CI	Typed length
HR expense	10	113	**0.0885**	0.049–0.155	35–44; median 42
Finance invoice	45	76	**0.592**	0.480–0.696	—
Finance managerial	27	32	**0.844**	0.683–0.931	38–45; median 40
HR onboarding	61	55	1.109	0.935–1.000	27–30; median 29
HR benefits	18	34	0.529	0.367–0.686	33–38; median 36

Table 43. Observed document-switching and comment-length budget. Variation is itself a prioritisation signal.

20.3 Counterfactual — what the prototype removes

Avoided	Amount	Derivation
Application switches	**6–20**	120 dispositions times measured rate and confidence interval
Keystrokes	**3,497**	observed comments contain 35–44 characters; the draft supplies 31
Field look-ups	**887**	record fields, regulation and clause collected in one view
Dispositions taken	**0**	unchanged by design

Table 44. Measured HR-expense counterfactual over 120 records. Rates remain ranges and are never converted to seconds without client inputs.

20.4 Scenario arithmetic — the model, not a number

The calculator ships with three inputs empty because they are the client's numbers. Its formula is:

$$H_{year} = \frac{a_{avoided}\, t_{action}\, n_{year}}{3600},$$
$$FTE_{equivalent} = \frac{H_{year}}{H_{productive}}.$$

Input	Source	Available?
actions_avoided_per_disposition	measured range in Table 44	yes
seconds_per_action	client environment; capture time compressed	no
dispositions_per_year	client volumes; recorded unavailable	no
annual_productive_hours	client HR policy	no
loaded_hourly_cost	client finance function	no

Table 45. Inputs to the scenario calculator. Missing quantities are left visibly empty rather than estimated.

Worked example — illustrative arithmetic only

The following assumptions are **not from this corpus**: 1.5 seconds per action, 20,000 annual dispositions and 1,800 productive hours per person. With the measured factor of roughly 29 keystrokes, 0.0885 switches and 7.4 look-ups avoided per disposition,

$$(29 + 0.0885 + 7.4) \times 1.5 \times 20{,}000/3600 \approx 304 \text{ hours/year} \approx 0.17 \text{ FTE}.$$

> Every factor except the first is an assumption. Changing 1.5 seconds to 0.5 changes the result threefold; inventing that input would be misleading.

20.5 The impact that is not in the arithmetic

Two effects are plausible and unmeasured. First, surfacing authority exceptions may avoid errors, but no incorrect disposition was observed and no error rate exists. Second, a new operator may benefit from seeing the governing clause, but onboarding and consistency were not measured.

20.6 What would make impact measurable

A one-queue pilot can collect four shipped metrics without a negative class: document override rate, share of drafts accepted unchanged, share of flags the operator agrees with, and actions per disposition before and after. That residual — not an argument — should decide whether a learned retriever is warranted.

21 Failure modes and limitations

Table 46. Limitations are reported with their mitigations and remaining gaps, not converted into a confidence score.

Risk	Consequence	Mitigation	Remaining gap
No production ground truth	every claim is inferential	independent review, three-key triangulation, 42-claim audit and declared permutation floors	corroborated, never proven
Reference annotations imperfect	reported scores are bounded	measured ≈ 176 ms lead, 2 s tolerance curve, truncated session retained	systematic bias may remain invisible
Screen text covers 4.59%	bounds every Layer 3 measure	disclosed wherever used; product reads a live interface	other 95% unknown
Values redacted	standard automatability index unavailable	redaction-safe structural proxy, explicitly labelled	proxy not validated against real index
Pixel-only evidence	some evidence cannot be parsed	screen text made vision unnecessary; fail-safe interface retained	pixel-only facts remain unread
Capture hash as identity	15 identifiers carry one clause	regulation-name plus hash; unknown hash refuses; siblings shown	no version ordering
Wrong document	operator may see wrong rule	64.2% and interval reported; MISS, alternatives and override log	confident error possible
Selector drift	panel may not find record	pinned identifiers and loud failure; adaptive recovery rejected	production stability unverified

Risk	Consequence	Mitigation	Remaining gap
Ambiguous rule	two readings	declared band semantics; overlap retained as a note	domain judgement
Decision supervision absent	autonomous disposition cannot be evaluated	product does not attempt it	revisit if real data contains rejections
Declared authority ceiling	one policy line moves OK/FLAG split	tagged as an inference after measuring that corpus cannot resolve it	client input required
Replica, not production	demonstration may diverge	real routes, IDs, fields and records; permanent honesty labels	no production-run claim
Three of twelve queues	limited coverage	shared definitions make missing modules visible	extension effort unvalidated
No production volume	total impact unavailable	measured factor and formula handed over; empty calculator	client must supply three numbers
No causal impact measure	product's strongest premise may be wrong	limitation and resolving experiment are first-class	see finding below
Rejection never observed	unseen workflow branch	rendered and labelled as never observed	unknowable here
Identical footprint, different purpose	hard limit of UI telemetry	named as a finding	no clustering method removes it

The headline limitation, stated plainly

The evidence establishes sequence, not causation. Operators open a governing document before disposing far above chance in 10 of 11 modules, most below $p < 0.001$. But the other failed gates suggest they may not be deciding from it. If document-open is procedural, habitual or an artifact of window placement, surfacing it faster saves keystrokes without changing the work. Only an operator interview or an A/B test with the document suppressed can settle this. That experiment should be run first.

22 Security, privacy and governance

Table 47. Security, privacy and governance position.

Concern	Position
Personal data	records concern named employees and vendors. Source values are redacted at capture and the product reads only the committed evidence corpus
Data residency	**no data leaves the machine**: no API key, model endpoint, network call or model download; the challenger is local and source is tested for network, credential and model-loading imports
Screenshots	4,746 contain business data; **zero were sent anywhere**. The vision pass was cancelled after measurement: 0 images, 0 calls, $0.00

Concern	Position
Clipboard	clipboard events are boundary evidence; clipboard content is not persisted or read
Document content	regulations remain local and are quoted verbatim for provenance. Clause text is never machine-translated at runtime; English glosses are non-authoritative
Trace and audit	record identifiers, suggestions and human responses are deliberately logged in a local, append-only, hash-chained and independently verifiable trail
Model inputs	there are none at runtime; no request path reaches a model
Access boundaries	evidence corpus is read-only by test; the frozen inference layer is not importable from product write paths
Human approval	irreversible action remains human; no code path performs it and a source-reading test enforces the boundary
Auditability	every statement resolves to document, clause, character span, capture and session
Document versioning	weakest area: hashes have no version ordering. Unknown hashes refuse rather than guess; production needs an authoritative register
Unowned automation	each rule is a reviewable line in a declaration beside its evidence, addressing the “stray robot” governance problem
Secrets	none. One optional environment variable enables the local challenger; no key is required

> **Governance is structural, not a policy appendix**
>
> The evidence corpus is read-only, every assertion carries provenance, every human response is auditable, and the only irreversible action is absent from the product’s capabilities. These are implementation properties verified by tests.

23 What we would do next

Each item follows from a measured limitation rather than from a wish list.

23.1 Immediate production hardening

1. **Create an authoritative document register with version ordering.** The system can identify a capture but cannot establish whether it is current.
2. **Promote selector declarations into a versioned startup check.** Interface drift must fail before work begins.
3. **Resolve the authority ceiling with the client.** It is the only declared policy parameter and directly moves the OK/FLAG split.
4. **Confirm the module key against a live system.** URL disagrees with the observed screen in 21% of captures; the product uses screen header and record prefix instead.

23.2 Additional evaluation

5. **Recover the 148 labelled retrieval pairs** so the challenger can be scored against the registered 71.5% bar.
6. **Run the causal experiment**: an operator interview or A/B test with the document suppressed to determine whether document-open is causal.
7. **Pilot one queue and collect four shipped metrics**: document override, draft acceptance, flag agreement and actions per disposition.
8. **Measure clause localisation** against a small human-labelled set; it is currently tested but not scored.

23.3 Broader coverage

9. **Add the next module through a declaration and measure the effort.** That is the honest test of whether the foundation generalises.
10. **Resolve the purchase-order warning marker.** One answer unblocks a candidate currently recorded as unevaluable.

23.4 Model and retrieval improvements, each with its trigger

11. **Try metric learning from bounded model-generated execution-pair labels.** The probe's 0.891 accuracy shows an objective gap rather than a missing representation; fit view weights and ablate against the current rule.
12. **Promote a learned retriever only after it clears the registered bar**: pooled top-1 above 71.5%, or a top-3 result the panel can genuinely use.
13. **Add clause-level per-record conditioning** on amount band, expense category or vendor class — information a static lookup cannot express.

23.5 Integration

14. **Establish whether bulk import or an API exists.** If staff are hand-keying despite an existing path, the correct answer includes process change, not only software.
15. **Place a deterministic executor behind the same declarations** only after a rejection branch has been observed and false-accept rate can be measured. The declarations are ready; the licensing evidence is not.

24 Conclusion

What was built. A four-layer system: executions recovered from raw desktop telemetry at boundary F1 0.690, 2.67× the strongest baseline and a segment-count ratio of 0.98; recurring processes discovered without labels at ARI 0.681 over 17 classes; the production dataset analysed into a twelve-module inventory; and ProcAssist, a running document-grounded assistant that prepares back-office approvals while leaving the decision to a human.

What was learned. Recorder artifacts are not business boundaries, and the apparent unit of work was hiding a third of the boundary evidence inside itself. Process identity without labels is a similarity-and-discovery problem. A linear probe can show that information is present rather than missing. A technical route is not a business process, and only validation against the interface the human saw catches that error. The most frequent process can be the least automatable, so a composite score would conceal the important finding. Finally, an outcome column containing only successes is not evidence that work is easy; it is evidence that the intended supervision does not exist there.

Why the architecture ended up this way. Every boundary was drawn by measurement. A language model was sized five times: against ceilings of ARI 0.478, +0.026 F1 and a probe accuracy of 0.891; against a document lookup at 64.2% with a 71.5% registered bar; and against a comment template matched by 142 of 223 observed strings. It was adopted once, at design time, where its artifact rather than the model executes. Unsupervised learning does the work no rule could do. Retrieval provides provenance. Deterministic validation carries invariants. The irreversible judgement remains human because 0 of 225 dispositions were negative and a system that cannot be evaluated on its errors must not make them.

What supports the product. Operators open a governing document before disposing far above a permutation null in 10 of 11 modules, most below $p < 0.001$. Retrieval is measured with a confidence interval and a bar fixed in advance. Impact is measured in the one unit the data supports; unavailable inputs remain empty. The system runs, 449 tests pass, and the boundary against autonomous disposition is enforced by a test that reads the source.

What remains uncertain. Whether document-open is causal; whether the challenger beats the lookup; whether a native integration path changes the right intervention; whether the declared authority ceiling matches client policy; and whether faster approval is better approval. The shipped instrumentation and a pilot can answer those questions; this corpus cannot.

The practical value

On one queue, unattended, ProcAssist resolves **23 governing documents**, stages **120 drafts**, and avoids a measured **6–20 application switches**, **3,497 keystrokes** and **887 look-ups** while taking **zero dispositions**. That final number is the product: everything the evidence licenses is prepared, and the one action it does not license remains deliberately and verifiably with the accountable person.

25 References

[1] **PROCESS AND TASK MINING**

Leno, V., Polyvyanyy, A., Dumas, M., La Rosa, M., Maggi, F.M. (2021). *Robotic Process Mining: Vision and Challenges.* Business & Information Systems Engineering 63(3). https://link.springer.com/article/10.1007/s12599-020-00641-4

[2] Bosco, A., Augusto, A., Dumas, M., La Rosa, M., Fortino, G. (2019). *Discovering Automatable Routines from User Interaction Logs.* BPM Forum. https://lepo.it.da.ut.ee/~dumas/pubs/rpadiscovery.pdf

[3] UiPath. *Task Mining — Unassisted Task Mining analysis guide.* https://docs.uipath.com/task-mining/automation-cloud/latest/user-guide/unassisted-task-mining-analysis-guide

[4] **AUTOMATION CANDIDATE SELECTION AND PRIORITISATION**

Wanner, J. et al. (2019). *Process Selection in RPA Projects — Towards a Quantifiable Method of Decision Making.* ICIS 2019. https://aisel.aisnet.org/icis2019/business_models/business_models/6/

[5] *Digging for Gold in RPA Projects — A Quantifiable Method to Identify and Prioritize Suitable RPA Process Candidates.* CAiSE 2021. https://link.springer.com/chapter/10.1007/978-3-030-79382-1_19

[6] *A method for selecting processes for automation with AHP and TOPSIS.* Heliyon (2023). https://www.sciencedirect.com/science/article/pii/S2405844023008903

[7] UiPath. *Automation Hub — Information About the Idea Score.* https://docs.uipath.com/automation-hub/automation-cloud/latest/user-guide/information-about-the-idea-score

[8] **ROBOTIC PROCESS AUTOMATION: THEMES, MAINTAINABILITY AND CONTROL**

Robotic Process Automation in Business Process Management: A Systematic Literature Review and an Integrated Framework. Technologies 14(4), 225. https://www.mdpi.com/2227-7080/14/4/225

[9] *How to Keep RPA Maintainable?* BPM 2020. https://www.researchgate.net/publication/344176607_How_to_Keep_RPA_Maintainable

[10] Syed, R. et al. (2020). *Robotic Process Automation: Contemporary Themes and Challenges.* Computers in Industry.

[11] UiPath. *Studio — About Object Repository.* https://docs.uipath.com/studio/standalone/2023.4/user-guide/about-object-repository

[12] Protiviti. *Building Bot Boundaries: RPA Controls in SOX Systems.* https://blog.protiviti.com/2018/10/31/building-bot-boundaries-rpa-controls-in-sox-systems/

[13] **AUTONOMOUS AGENTS AND HUMAN-IN-THE-LOOP ARCHITECTURE**

Drouin, A. et al. *WorkArena: How Capable Are Web Agents at Solving Common Knowledge Work Tasks?* ServiceNow Research. https://servicenow.github.io/WorkArena/

[14] AWS. *Customize business rules for intelligent document processing with human review and BI visualization.* https://aws.amazon.com/blogs/machine-learning/customize-business-rules-for-intelligent-document-processing-with-human-review-and-bi-visualization/

[15] *Saga pattern made easy* — compensating transactions and idempotency in distributed workflows. https://temporal.io/blog/saga-pattern-made-easy

[16] **JAPANESE ENTERPRISE AND PUBLIC-SECTOR CONTEXT**

総務省 (Ministry of Internal Affairs and Communications). 自治体におけるRPA導入ガイドブック — RPA introduction guidebook for local government. https://www.soumu.go.jp/main_content/000731625.pdf

[17] 三菱総合研究所 (Mitsubishi Research Institute). RPAを「野良ロボ」にしない — on avoiding unowned automations. https://www.mri.co.jp/knowledge/mreview/201803-5.html

[18] IPA. DX動向2025 — digital-transformation trends. Cited for existence; figures not extracted. https://www.ipa.go.jp/digital/chousa/dx-trend/dx-trend-2025.html

[19] 日立ソリューションズ. 部門別RPA活用事例 — RPA use cases by department. https://www.hitachi-solutions.co.jp/rpa/column/rpa_vol13.html

[20] 給与計算と会計の連携とは？二重入力を減らす進め方と確認ポイント — on payroll-to-accounting integration and reducing duplicate entry. https://bakuraku.jp/knowledge/knowledge-payroll/what-is-the-integration-of-payroll-and-accounting/

[21] 会計システム連携とは｜販売・経費・給与とつなぐ方法と注意点 — accounting-system integration methods and caveats. https://www.asteria.com/jp/warp/blog/165665.html

[22] **LABOUR-MARKET CONTEXT** (cited as motivation, not as a source of any figure used)

Bank of Japan (2025). *Japan's Labor Market under Demographic Decline.* https://www.boj.or.jp/en/about/press/koen_2025/data/ko250824a1.pdf

[23] IMF Working Paper 2025/184. *The Impact of Aging and AI on Japan's Labor Market.* https://www.elibrary.imf.org/view/journals/001/2025/184/article-A001-en.xml

[24] JILPT. *Japan Labor Issues*, vol. 9. https://www.jil.go.jp/english/jli/documents/2025/053-00.pdf

A machine-readable BibTeX version of this list is committed alongside the report source as references.bib. No bibliographic detail appears in either that is not present in the source report; entries that the source gives only by title, venue and URL are recorded with those fields alone.

A Dataset statistics

Table 48. Dataset populations.

Property	Dataset A	Dataset B

Sessions	63	15
Recording chunks	133	20
Events	162,768	20,477
Screenshots	34,563	4,746
Distinct machines	8	4
Ground-truth files	63	0
Reference documents	—	13 regulations; 112 captures; 418 clauses (365 retrievable at 12 or more characters)
Extracted screen text	—	939 events (4.59%)
Records extracted	—	3,011 across 13 prefixes
Outcomes extracted	—	549, including 225 positive toasts
Queue tables	—	216
Business modules	—	12; one absent from the original route map
Characters swept for outcome variance	—	25,495,968

B Metric definitions

Metric	Definition	Direction
Boundary F1 at τ	precision/recall harmonic mean on boundary positions matched within tolerance τ	higher
Segment-count ratio	produced segments divided by true segments; must accompany boundary F1	1.00 ideal
Reachability	share of true boundaries the candidate space can nominate; not an accuracy measure	higher
Segmental F1 at IoU 0.5	whole-segment agreement at 50% overlap	higher
WindowDiff / Pk	sliding-window segmentation disagreement	lower
Adjusted Rand Index	chance-corrected partition agreement	higher
Normalised Mutual Information	shared information between partitions	higher
Homogeneity	whether each cluster contains only one true class	higher
Completeness	whether each true class lands in one cluster	higher
Match rate	share of produced segments matched to truth; matched subsets with different rates are not comparable	context
Event ARI	identity agreement over all events	higher
Interruption stitching	interrupted executions whose fragments share a label; rises mechanically with coarser labels	contextual
Wilson 95% interval	binomial interval used for action-budget rates	—

C Baseline comparison

Six baselines were evaluated leave-one-session-out across 63 sessions. The delivered pipeline reaches boundary F1 0.690 at count ratio 0.98, versus the strongest baseline's 0.258 at 2.83, and leads on event ARI 0.297 versus 0.250.

D Process-discovery experiments

Arm	ARI	NMI	Homogeneity	Completeness	Labels
Structural signature	0.431	0.563	0.622	0.514	51
kNN graph + communities	0.583	0.688	0.679	0.698	15
+ saturation tie-break	**0.681**	**0.738**	**0.751**	**0.725**	**17**

Table 50. Process-discovery ablation.

Rejected variants include inverse-document-frequency reweighting (separability Cohen's d 1.535 $\rightarrow$ 1.508; ARI -0.031) and density clustering at a tuned radius (stitching 0.857 at ARI -0.001). Language-model ceilings were ARI 0.478 for perfect label merging and $+0.026$ F1 for perfect boundary adjudication; the supervised probe reached 0.891 accuracy / ARI 0.786.

E Step 2 candidate analysis

The full inventory carries 37 measured columns per module. Business volume, monetary value and error rate are recorded as UNAVAILABLE because telemetry cannot recover them. The main report retains frequency and automatability as separate axes rather than hiding this limitation in a score.

F Hypothesis matrix

Forty-four tests cover 11 modules and four hypotheses. Nulls use 10,000 draws, seed 7 and a pre-declared evidence floor of $n \geq 8$.

Hypothesis	Sup.	Ref.	Untest.	Insuff.	Representative statistic
Rule predicts disposition	0	0	8	3	33/33 positive; dependent variable constant
Difficulty predicts effort	0	3	0	8	$\rho = -0.2737$, $p = 0.2317$; null median $\lvert\rho\rvert = 0.1564$
Document before disposition	10	1	0	0	0.785 observed versus 0.492 null median, $p = 0.0$
Record carries rule variables	0	4	0	7	amount on 12/133 (9.02%); authority on 0

Table 51. Layer 3 hypothesis matrix.

G API schema

The service exposes nine API endpoints plus four server-rendered pages. The result object is immutable, forbids extra fields, and raises if a draft ever carries a judgement. The endpoint table and lifecycle appear in § 14.

H Example request and response

```
// POST /v1/prepare
{ "record_id": "P1-07106250-010", "trace": true }
{
  "definition_id": "hr_expense_settlement",
  "module_id": "M01",
  "record_id": "P1-07106250-010",
  "record_fields": { "区分": "接待交際費", "金額": "円135,181" },
  "status": "FLAG",
  "document": {
    "regulation_name_ja": "接待交際費規程",
    "document_id": "3ece1f98df11...",
    "identity_note_en": "15 captures under 5 window titles"
  },
  "clause": {
    "clause_id": "c002",
    "heading_ja": "第 2 条（承認権限）",
    "char_span": [137, 189]
  },
  "checks": [{
    "status": "FLAG",
    "expected_ja": "社長承認",
    "observed_ja": "役員承認（このキューの上限）",
    "semantics": "MOST_SPECIFIC_TIER",
    "basis": "INF"
  }],
  "draft": {
    "text_ja": "経費精算確認済み。費目：接待交際費　金額：円。。135,181",
    "judgement_slot": "judgement",
    "judgement": null,
    "observed_n": 142
  },
  "population": { "same": 12, "of": 120 },
  "trace": { "total_ms": 2.4 }
}
```

The omitted fields carry full expected/observed/why triplets and provenance. The critical invariant is visible: `"judgement": null`.

I Configuration

There are no secrets, keys, model endpoints or database servers. One optional environment variable enables the off-by-default retrieval challenger. The substantive configuration is three declaration files; their only policy knob is the per-queue authority ceiling, tagged as an inference wherever it appears.

J End-to-end run

See § 18: two commands start the system; the walkthrough covers all four statuses, the audit reader and test commands. The result is reproducible without network access or model services.

K Impact calculation methodology

The measured input is per-disposition action counts over 20,477 events, with Wilson intervals, and the 120-record replica counterfactual. Seconds per action, annual volume, productive hours and loaded cost must come from the client. The formula and illustrative example are in § 20; three assumed values are explicitly separated from corpus measurements.

L Supporting material in the repository

The repository contains typeset analysis artifacts, the evidence corpus behind each claim, an independent review, a 42-claim audit, a per-session work log, an engagement narrative, structured commit history and the prompts used for generative-AI assistance. The reading order is in README_PROJECT.md.